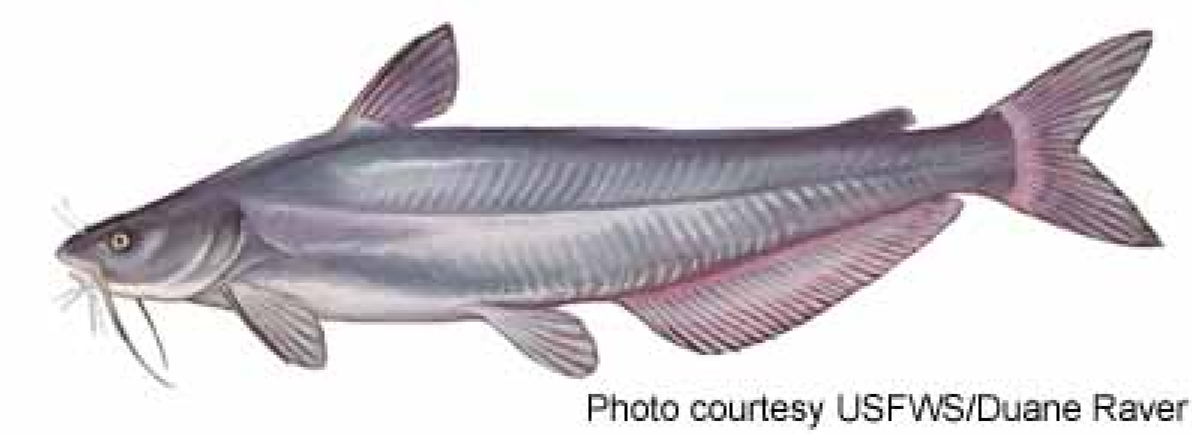
BIOL 6220 Final Project - Cambria Miller



Blue Catfish *Ictalurus furcatus*

Objective: This project focuses on identifying areas where blue catfish *Ictalurus furcatus* are most likely to establish and spread in the Southeastern United States. This approach uses **Species Distribution Modeling (SDM),** comparing two models to predict habitat suitability for blue catfish based on environmental variables

Methodology:

Occurrence data for blue catfish retrieved from the Global Biodiversity Information Facility (GBIF)

Environmental layers obtained to model potential habitats for the species

Models

Generalized Linear Model (GLM)

Logistic Regression Model

```r
# Set working drive
setwd("/Users/millerca23/Library/CloudStorage/OneDrive-EastCarolinaUniversity/Fall 2024/BIOL 6220 Practical Computing/Finalprojectpt3")
```

```r
# Load Packages
library(sf)
library(raster)
library(sdm)
library(ggplot2)
library(ggmap)
library(rnaturalearth)
library(rnaturalearthdata)
library(rgbif)
library(terra)
library(geodata)
library(sdmpredictors)
library(stringr)
library(ggspatial)
library(stars)
library(elevatr)
library(tigris)
library(ggsflabel)
library(MIAmaxent)
library(data.table)
library(predicts)
library(dplyr)
library(tibble)
library(tidyterra)
library(tidyverse)
```

```
# Retrieve occurrence data for blue catfish from GBIF fo
r Southeastern US

Ictalurus.furcatus_GBIF <- occ_data(scientificName = "Ic
talurus furcatus", hasCoordinate = TRUE, decimalLongitud
e = "-100, -60", decimalLatitude = "0, 50", limit = 1000
0)

Ictalurus.furcatus_GBIF <- as.data.frame(Ictalurus.furca
tus_GBIF$data)
```

Obtain Environmental Layers

```
# View available environmental layers (filtered for fres
hwater and Bioclim layers)
datasets <- list_datasets(terrestrial = TRUE, marine =
F, freshwater = TRUE)
layers <- list_layers(datasets)

(fw_layers <- layers %>%
  filter(freshwater == "TRUE") %>%
  filter(str_detect(name, "^Bioclim")) %>%
  filter(str_detect(layer_code, "_avg")))
```

```
##    dataset_code      layer_code       name descriptio
n terrestrial marine
## 1    Freshwater FW_hydro_avg_01  Bioclim 1   Bioclim 
1       FALSE  FALSE
## 2    Freshwater FW_hydro_avg_02  Bioclim 2   Bioclim 
2       FALSE  FALSE
## 3    Freshwater FW_hydro_avg_03  Bioclim 3   Bioclim 
3       FALSE  FALSE
## 4    Freshwater FW_hydro_avg_04  Bioclim 4   Bioclim 
4       FALSE  FALSE
## 5    Freshwater FW_hydro_avg_05  Bioclim 5   Bioclim 
5       FALSE  FALSE
## 6    Freshwater FW_hydro_avg_06  Bioclim 6   Bioclim 
6       FALSE  FALSE
## 7    Freshwater FW_hydro_avg_07  Bioclim 7   Bioclim 
7       FALSE  FALSE
## 8    Freshwater FW_hydro_avg_08  Bioclim 8   Bioclim 
8       FALSE  FALSE
## 9    Freshwater FW_hydro_avg_09  Bioclim 9   Bioclim 
9       FALSE  FALSE
## 10   Freshwater FW_hydro_avg_10 Bioclim 10  Bioclim 1
0       FALSE  FALSE
## 11   Freshwater FW_hydro_avg_11 Bioclim 11  Bioclim 1
1       FALSE  FALSE
## 12   Freshwater FW_hydro_avg_12 Bioclim 12  Bioclim 1
2       FALSE  FALSE
## 13   Freshwater FW_hydro_avg_13 Bioclim 13  Bioclim 1
3       FALSE  FALSE
## 14   Freshwater FW_hydro_avg_14 Bioclim 14  Bioclim 1
4       FALSE  FALSE
## 15   Freshwater FW_hydro_avg_15 Bioclim 15  Bioclim 1
5       FALSE  FALSE
## 16   Freshwater FW_hydro_avg_16 Bioclim 16  Bioclim 1
6       FALSE  FALSE
```

```
## 17   Freshwater FW_hydro_avg_17 Bioclim 17  Bioclim 1
7       FALSE  FALSE
## 18   Freshwater FW_hydro_avg_18 Bioclim 18  Bioclim 1
8       FALSE  FALSE
## 19   Freshwater FW_hydro_avg_19 Bioclim 19  Bioclim 1
9       FALSE  FALSE
##    freshwater cellsize_equalarea cellsize_lonlat
units
## 1        TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 2        TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 3        TRUE                815     0.008333333
* 100
## 4        TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 5        TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 6        TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 7        TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 8        TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 9        TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 10       TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 11       TRUE                815     0.008333333 [deg
rees Celsius] * 10
## 12       TRUE                815     0.008333333
[mm]
## 13       TRUE                815     0.008333333
[mm]
## 14       TRUE                815     0.008333333
```

```
[mm]
## 15       TRUE                815    0.008333333
* 100
## 16       TRUE                815    0.008333333
[mm]
## 17       TRUE                815    0.008333333
[mm]
## 18       TRUE                815    0.008333333
[mm]
## 19       TRUE                815    0.008333333
[mm]
##            primary_type primary_spatial_resolution
## 1  in situ measurement                 30 arsecond
## 2  in situ measurement                 30 arsecond
## 3  in situ measurement                 30 arsecond
## 4  in situ measurement                 30 arsecond
## 5  in situ measurement                 30 arsecond
## 6  in situ measurement                 30 arsecond
## 7  in situ measurement                 30 arsecond
## 8  in situ measurement                 30 arsecond
## 9  in situ measurement                 30 arsecond
## 10 in situ measurement                 30 arsecond
## 11 in situ measurement                 30 arsecond
## 12 in situ measurement                 30 arsecond
## 13 in situ measurement                 30 arsecond
## 14 in situ measurement                 30 arsecond
## 15 in situ measurement                 30 arsecond
## 16 in situ measurement                 30 arsecond
## 17 in situ measurement                 30 arsecond
## 18 in situ measurement                 30 arsecond
## 19 in situ measurement                 30 arsecond
##
primary_source
## 1  WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
```

```
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 2  WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 3  WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 4  WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 5  WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 6  WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 7  WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 8  WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
```

```
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 9  WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 10 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 11 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 12 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 13 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 14 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965–1
978.
## 15 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
```

```
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965-1
978.
## 16 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965-1
978.
## 17 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965-1
978.
## 18 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965-1
978.
## 19 WorldClim, Hijmans, R. J., Cameron, S. E., Parra,
J. L., Jones, P. G., & Jarvis, A. (2005). Very high reso
lution interpolated climate surfaces for global land are
as. International Journal of Climatology, 25(15), 1965-1
978.
##    start_year start_month start_day end_year end_mont
h end_day derivation month
## 1        1950           1         1     2000
1       1    average    NA
## 2        1950           1         1     2000
1       1    average    NA
## 3        1950           1         1     2000
1       1    average    NA
## 4        1950           1         1     2000
1       1    average    NA
## 5        1950           1         1     2000
1       1    average    NA
```

```
## 6        1950          1          1     2000
1       1    average    NA
## 7        1950          1          1     2000
1       1    average    NA
## 8        1950          1          1     2000
1       1    average    NA
## 9        1950          1          1     2000
1       1    average    NA
## 10       1950          1          1     2000
1       1    average    NA
## 11       1950          1          1     2000
1       1    average    NA
## 12       1950          1          1     2000
1       1        sum    NA
## 13       1950          1          1     2000
1       1        sum    NA
## 14       1950          1          1     2000
1       1        sum    NA
## 15       1950          1          1     2000
1       1        sum    NA
## 16       1950          1          1     2000
1       1        sum    NA
## 17       1950          1          1     2000
1       1        sum    NA
## 18       1950          1          1     2000
1       1        sum    NA
## 19       1950          1          1     2000
1       1        sum    NA
##    is_surface version
## 1        TRUE      10
## 2        TRUE      10
## 3        TRUE      10
## 4        TRUE      10
## 5        TRUE      10
## 6        TRUE      10
```

```
## 7        TRUE      10
## 8        TRUE      10
## 9        TRUE      10
## 10       TRUE      10
## 11       TRUE      10
## 12       TRUE      10
## 13       TRUE      10
## 14       TRUE      10
## 15       TRUE      10
## 16       TRUE      10
## 17       TRUE      10
## 18       TRUE      10
## 19       TRUE      10
##
layer_url
## 1  https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_01_lonlat.tif
## 2  https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_02_lonlat.tif
## 3  https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_03_lonlat.tif
## 4  https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_04_lonlat.tif
## 5  https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_05_lonlat.tif
## 6  https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_06_lonlat.tif
## 7  https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_07_lonlat.tif
## 8  https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_08_lonlat.tif
## 9  https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_09_lonlat.tif
## 10 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_10_lonlat.tif
```

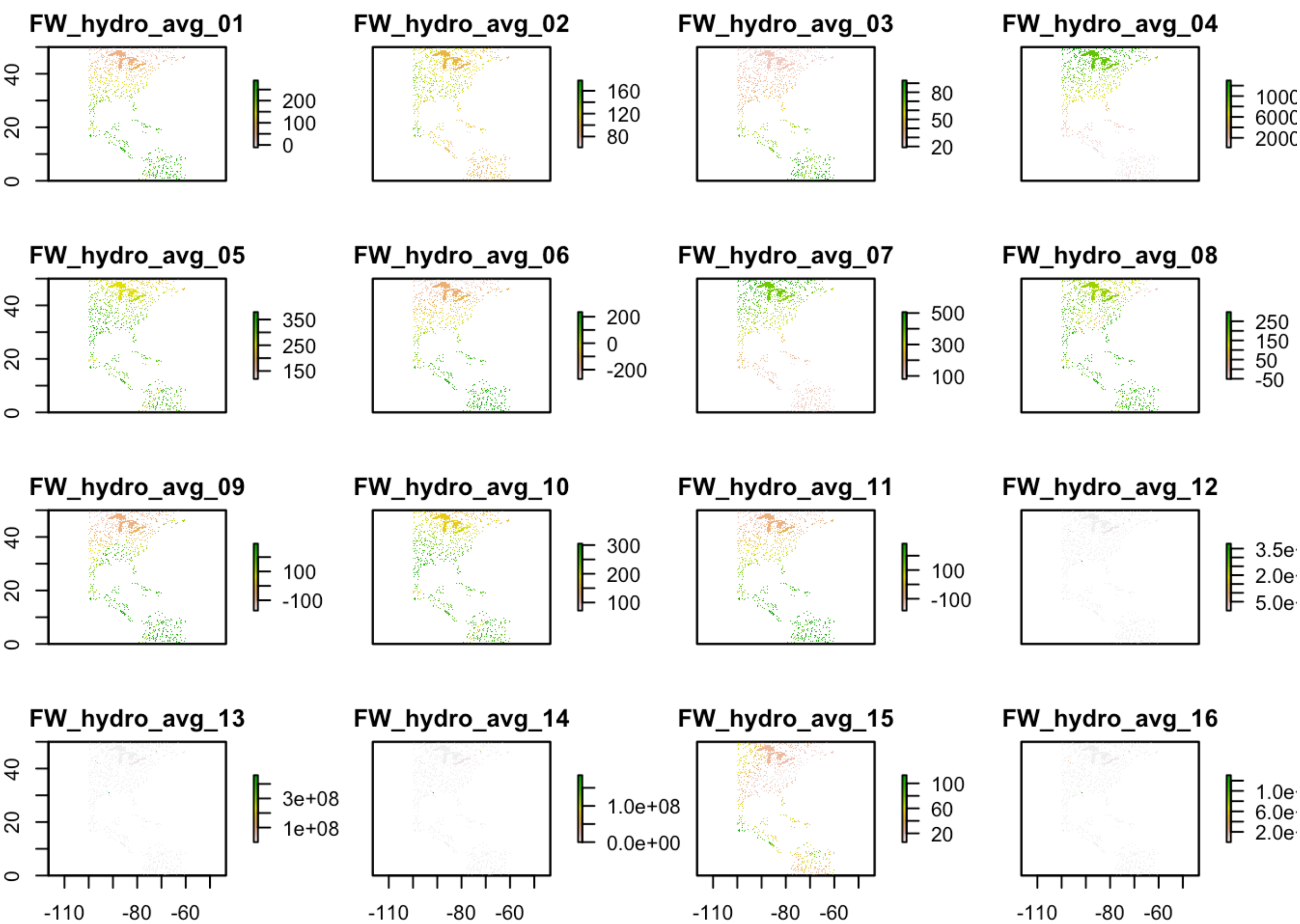
```
## 11 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_11_lonlat.tif
## 12 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_12_lonlat.tif
## 13 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_13_lonlat.tif
## 14 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_14_lonlat.tif
## 15 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_15_lonlat.tif
## 16 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_16_lonlat.tif
## 17 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_17_lonlat.tif
## 18 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_18_lonlat.tif
## 19 https://www.lifewatch.be/sdmpredictors/FW_hydro_av
g_19_lonlat.tif
```

```
fw_layers$layer_code
```

```
##  [1] "FW_hydro_avg_01" "FW_hydro_avg_02" "FW_hydro_av
g_03" "FW_hydro_avg_04"
##  [5] "FW_hydro_avg_05" "FW_hydro_avg_06" "FW_hydro_av
g_07" "FW_hydro_avg_08"
##  [9] "FW_hydro_avg_09" "FW_hydro_avg_10" "FW_hydro_av
g_11" "FW_hydro_avg_12"
## [13] "FW_hydro_avg_13" "FW_hydro_avg_14" "FW_hydro_av
g_15" "FW_hydro_avg_16"
## [17] "FW_hydro_avg_17" "FW_hydro_avg_18" "FW_hydro_av
g_19"
```

```
# Crop environmental layers to the geographic extent of occurrence data
layercodes <- fw_layers$layer_code
env <- load_layers(layercodes,  datadir = "sdm_data_layers")
```

```
easternUS <- raster::crop(env, raster::extent(-100, -60, 0, 50))
plot(easternUS)
```

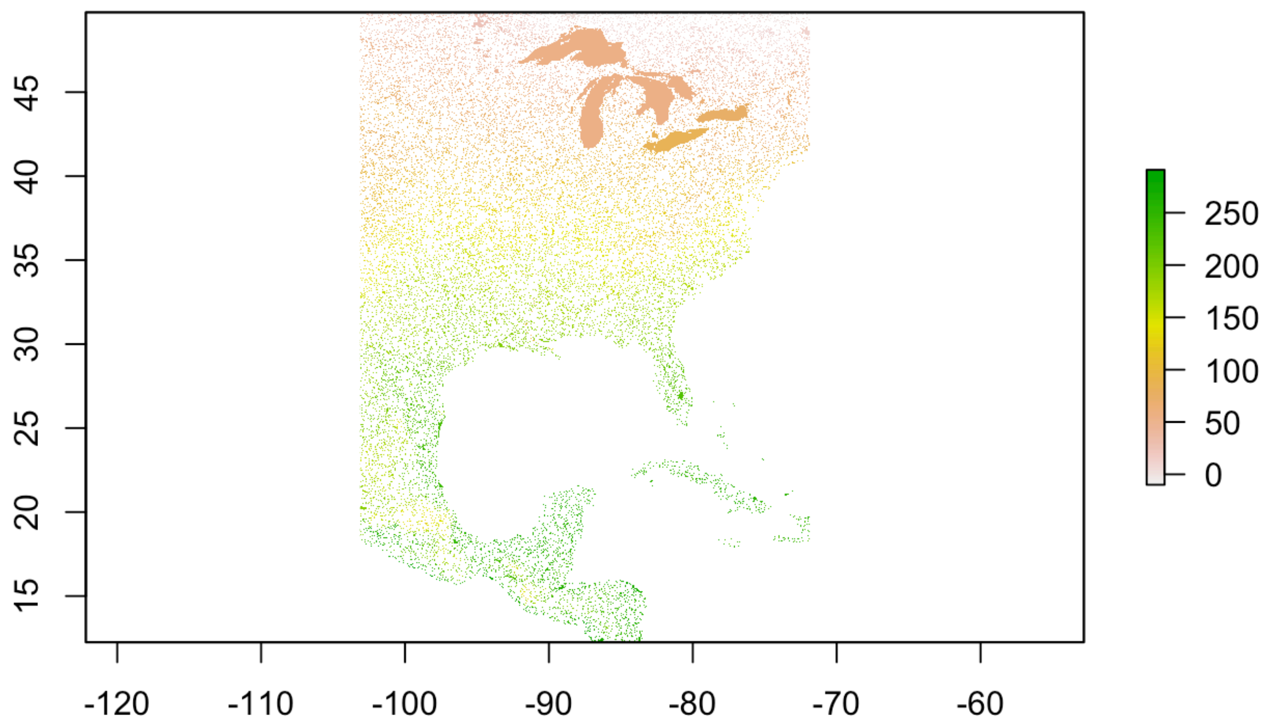


Choose occurence points for SDMs

```
# Select non-NA occurrence points
obs_data <- Ictalurus.furcatus_GBIF[!is.na(Ictalurus.fur
catus_GBIF$decimalLatitude), ]
```

Crop environmental layers to fit occurences

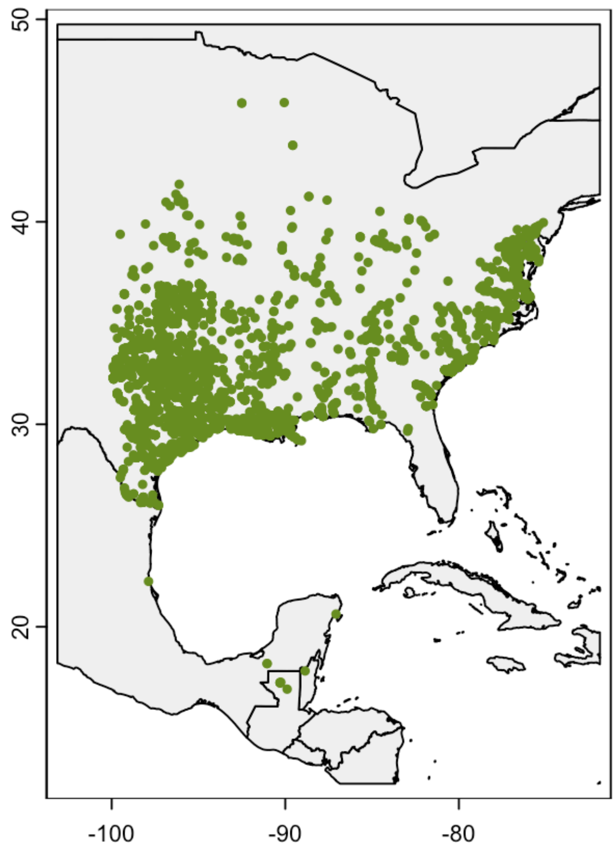
```
# Determine geographic extent of our data
max_lat <- ceiling(max(obs_data$decimalLatitude))
min_lat <- floor(min(obs_data$decimalLatitude))
max_lon <- ceiling(max(obs_data$decimalLongitude))
min_lon <- floor(min(obs_data$decimalLongitude))
# Store boundaries in a single extent object
geographic_extent <- raster::extent(x = c(min_lon, max_l
on, min_lat, max_lat))

# Make an extent that is 25% larger
sample_extent <- geographic_extent * 1.25

# Crop bioclim data to desired extent
bioclim_data <- crop(x = env, y = sample_extent)

# Plot the first of the bioclim variables to check on cr
opping
plot(bioclim_data[[1]])
```




```
bioclim_data_SR <- as(bioclim_data, "SpatRaster")
```

```
# Fetch occurrences
points <- cbind(obs_data$decimalLongitude, obs_data$decimalLatitude)
occfile <- tempfile(fileext = ".csv")
write.csv(cbind(points, value=1), occfile)
```

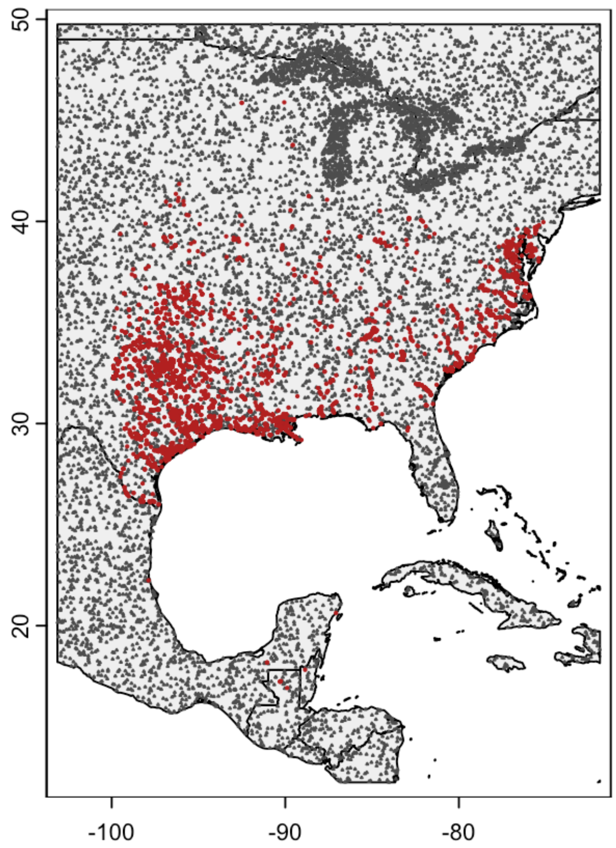
```
# Download data with geodata's world function to use for
our base map
world_map <- world(resolution = 3,
                   path = "data/")

# Crop the map to our area of interest
my_map <- crop(x = world_map, y = sample_extent)

# Plot the base map
plot(my_map,
     axes = TRUE,
     col = "grey95")

# Add the points for individual observations
points(x = obs_data$decimalLongitude,
       y = obs_data$decimalLatitude,
       col = "olivedrab",
       pch = 20,
       cex = 0.75)
```


50
40
30
20
-100
-90
-80

```
# Set the seed for the random-number generator to ensure
results are similar
set.seed(20210707)

# Randomly sample points (same number as our observed po
ints)
background <- spatSample(x = bioclim_data_SR,
                         size = 10000,    # generate 1,0
00 pseudo-absence points
                         values = FALSE, # don't need va
lues
                         na.rm = TRUE,   # don't sample
from ocean
                         xy = TRUE)      # just need coo
rdinates

# Look at first few rows of background
head(background)
```

```
##               x        y
## [1,]  -87.25417 32.09583
## [2,]  -80.90417 40.54583
## [3,]  -94.10417 41.43750
## [4,]  -82.91250 41.64583
## [5,]  -72.98750 49.10417
## [6,] -101.57917 21.89583
```

```
# Plot the base map
plot(my_map,
     axes = TRUE,
     col = "grey95")

# Add the background points
points(background,
       col = "grey30",
       pch = 2,
       cex = 0.1,)

# Add the points for individual observations
points(x = obs_data$decimalLongitude,
       y = obs_data$decimalLatitude,
       col = "firebrick",
       pch = 20,
       cex = 0.25)
```


50
40
30
20
-100
-90
-80

```
# Pull out coordinate columns, x (longitude) first, then
y (latitude) from presences
presence <- obs_data[, c("decimalLongitude", "decimalLat
itude")]
# Add column indicating presence
presence$pa <- 1

# Convert background data to a data frame
absence <- as.data.frame(background)
# Update column names so they match presence points
colnames(absence) <- c("decimalLongitude", "decimalLatit
ude")
# Add column indicating absence
absence$pa <- 0

# Join data into single data frame
all_points <- rbind(presence, absence)

#Convert column names to match bioclim
colnames(all_points) <- c("longitude", "latitude","pa")

# Reality check on data
head(all_points)
```

```
##   longitude latitude pa
## 1 -95.62070 35.46552  1
## 2 -90.04768 32.38832  1
## 3 -97.34791 27.68602  1
## 4 -95.09022 29.59558  1
## 5 -93.23460 35.30632  1
## 6 -97.87968 28.06486  1
```

```
# Extract climate data for occurrence points
bioclim_extract <- terra::extract(x = bioclim_data_SR,
                                  y = all_points[, c("longitud
e",
                                                      "latitud
e")],
                                  ID = FALSE)
```

```
# Combine occurrence data with climate data
points_climate <- cbind(all_points, bioclim_extract)

# Identify columns that are latitude & longitude
drop_cols <- which(colnames(points_climate) %in% c("long
itude", "latitude"))
drop_cols # print the values as a reality check
```

```
## [1] 1 2
```

```
# Remove the geographic coordinates from the data frame
points_climate <- points_climate[, -drop_cols]
```

```
# Create vector indicating fold
fold <- folds(x = points_climate,
              k = 5,
              by = points_climate$pa)

table(fold)
```

```
## fold
##    1    2    3    4    5
## 3904 3904 3905 3904 3904
```

```
testing <- points_climate[fold == 1, ]
training <- points_climate[fold != 1, ]
```

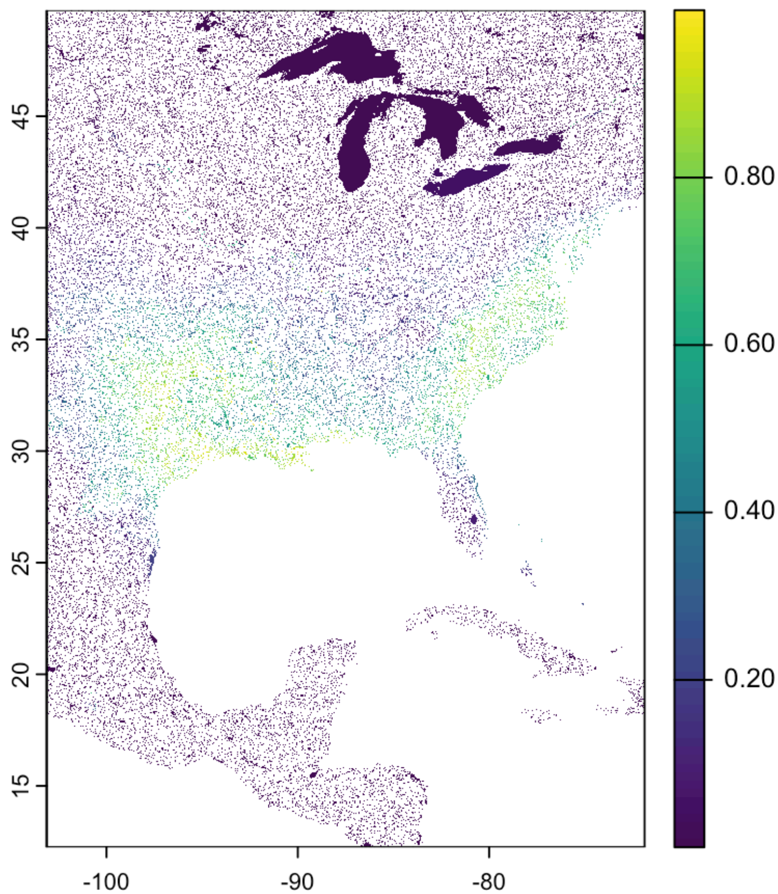
Species Distribution Model - GLM

```
# Build a model using training data
glm_model <- glm(pa ~ ., data = training, family = binom
ial())
```

```
# Predict habitat suitability using the GLM model
glm_predict <- predict(bioclim_data_SR, glm_model, type
= "response")
```

```
## |---------|---------|---------|---------|===========
=============================
```

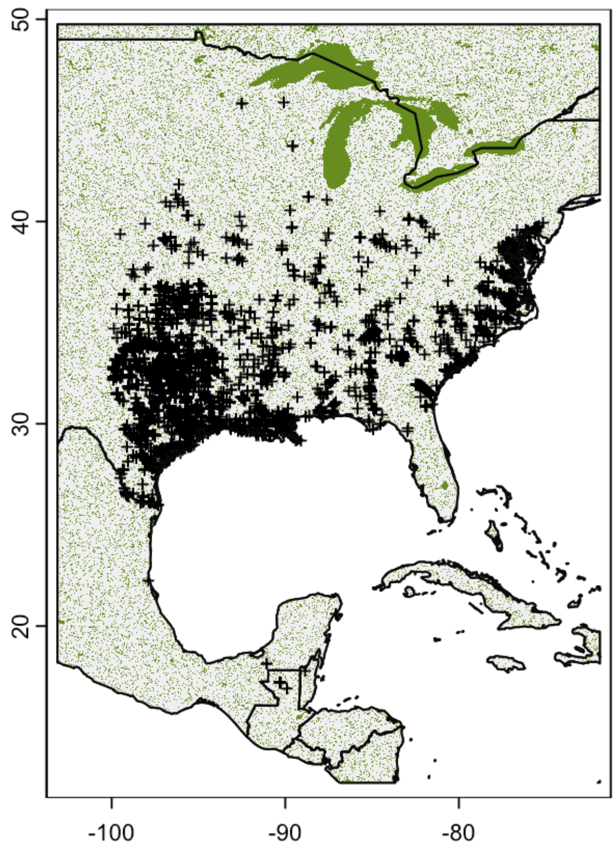
```
# Print GLM prediction
plot(glm_predict)
```




```
# Evaluate the GLM model using testing data
glm_eval <- pa_evaluate(p = testing[testing$pa == 1, ],
                        a = testing[testing$pa == 0, ],
                        model = glm_model,
                        type = "response")
```

```
# Determine minimum threshold for "presence"
glm_threshold <- glm_eval@thresholds$max_spec_sens
```

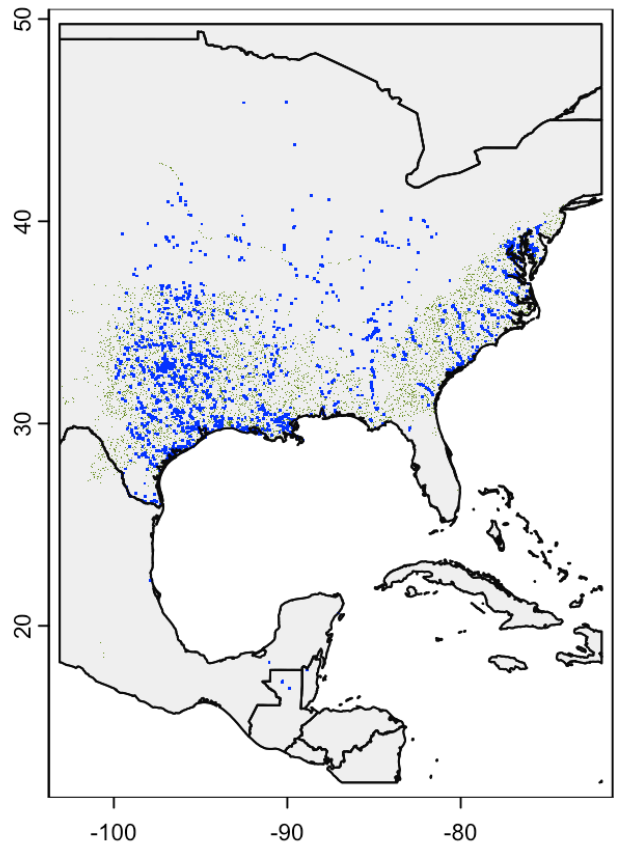
```
# Plot base map
plot(my_map,
     axes = TRUE,
     col = "grey95")

# Only plot areas where probability of occurrence is gre
ater than the threshold
plot(glm_predict > glm_threshold,
     add = TRUE,
     legend = FALSE,
     col = "olivedrab")

# And add those observations
points(x = obs_data$decimalLongitude,
       y = obs_data$decimalLatitude,
       col = "black",
       pch = "+",
       cex = 0.75)

# Redraw those country borders
plot(my_map, add = TRUE, border = "grey5")
```




```
glm_predict > glm_threshold
```

```
## class       : SpatRaster
## dimensions  : 4500, 3750, 1  (nrow, ncol, nlyr)
## resolution  : 0.008333333, 0.008333333  (x, y)
## extent      : -103.125, -71.875, 12.25, 49.75  (xmin,
xmax, ymin, ymax)
## coord. ref. : lon/lat WGS 84 (EPSG:4326)
## source(s)   : memory
## varname     : spat_52f1243ab7cf_21233
## name        :  lyr1
## min value   : FALSE
## max value   :  TRUE
```

```
# Plot base map
plot(my_map,
     axes = TRUE,
     col = "grey95")

# Only plot areas where probability of occurrence is greater than the threshold
plot(glm_predict > glm_threshold,
     add = TRUE,
     legend = FALSE,
     col = c(NA, "olivedrab")) # <-- Update the values HERE

# And add those observations
points(x = obs_data$decimalLongitude,
       y = obs_data$decimalLatitude,
       #col = alpha("blue",0.5),
       col = "blue",
       pch = ".",
       cex = 0.0000000000001
       )

# Redraw those country borders
plot(my_map, add = TRUE, border = "grey5")
```


50
40
30
20
-100
-90
-80

```
# Save PDF
pdf("ICT_GLM_Pred.pdf")

# Plot base map
plot(my_map,
     axes = TRUE,
     col = "grey95")

# Only plot areas where probability of occurrence is gre
ater than the threshold
plot(glm_predict > glm_threshold,
     add = TRUE,
     legend = FALSE,
     col = c(NA, "olivedrab")) # <-- Update the values H
ERE

# And add those observations
points(x = obs_data$decimalLongitude,
       y = obs_data$decimalLatitude,
       #col = alpha("blue",0.5),
       col = "blue",
       pch = ".",
       cex = 0.0000000000001
       )

# Redraw those country borders
plot(my_map, add = TRUE, border = "grey5")

dev.off()
```

```
## quartz_off_screen
##                 2
```

```
glm_model
```

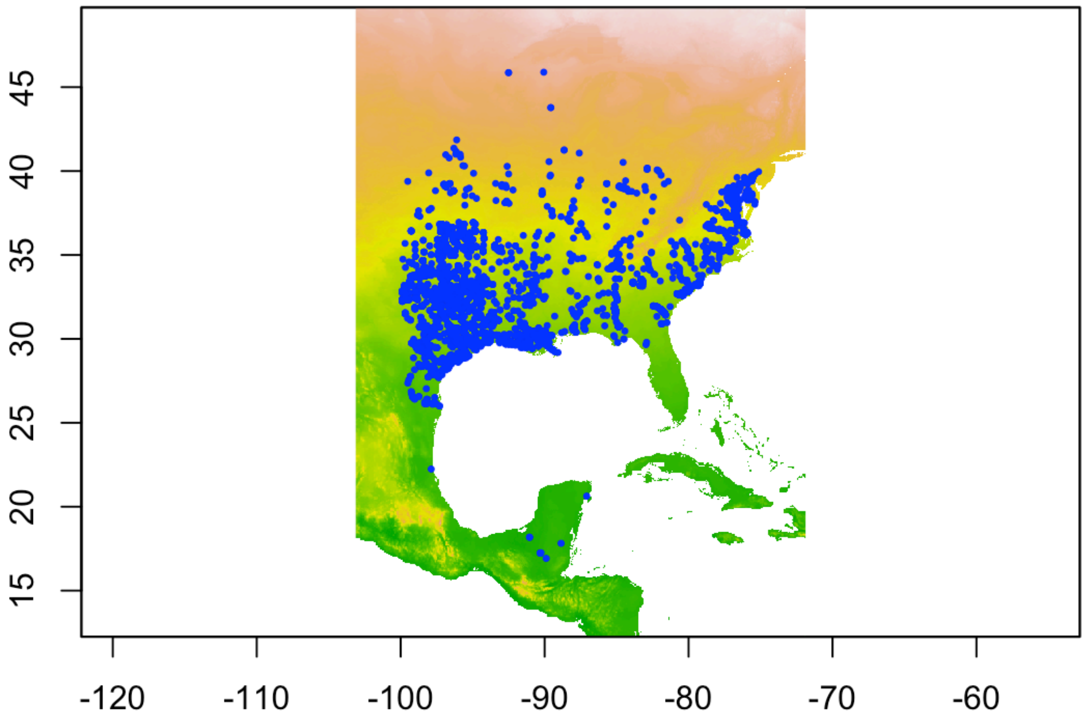
```
##
## Call:  glm(formula = pa ~ ., family = binomial(), dat
a = training)
##
## Coefficients:
##     (Intercept)  FW_hydro_avg_01  FW_hydro_avg_02  FW
_hydro_avg_03
##       9.805e+00       -6.040e-01        1.254e-01
-5.497e-01
## FW_hydro_avg_04  FW_hydro_avg_05  FW_hydro_avg_06  FW
_hydro_avg_07
##      -8.862e-03       -6.871e-02       -4.930e-03
NA
## FW_hydro_avg_08  FW_hydro_avg_09  FW_hydro_avg_10  FW
_hydro_avg_11
##       6.322e-03       -7.252e-03        7.059e-01
4.432e-02
## FW_hydro_avg_12  FW_hydro_avg_13  FW_hydro_avg_14  FW
_hydro_avg_15
##      -1.898e-08        7.050e-07       -6.838e-07
-8.701e-02
## FW_hydro_avg_16  FW_hydro_avg_17  FW_hydro_avg_18  FW
_hydro_avg_19
##      -2.159e-07        2.810e-07               NA
NA
##
## Degrees of Freedom: 10903 Total (i.e. Null);  10887 R
esidual
##   (4713 observations deleted due to missingness)
## Null Deviance:       13350
## Residual Deviance: 6073  AIC: 6107
```

```
geotiffs <- list.files(path = "data/climate/wc2.1_2.5
m/", pattern = ".tif")

dir.create("data/ascii_files")

for (i in geotiffs) {
  f <- raster::raster(paste0("data/climate/wc2.1_2.5m/",
i))
  r <- raster::readAll(f)
  r <- raster::crop(r, sample_extent)
  writeRaster(r, paste0("data/ascii_files/",i,".asc"), f
ormat = "ascii", overwrite=TRUE)
}
```

```
# Prepare occurrence data for MaxEnt model
GBIF4MIAmaxent <- as.data.frame(cbind(Ictalurus.furcatus
_GBIF$decimalLongitude, Ictalurus.furcatus_GBIF$decimalL
atitude))

GBIF4MIAmaxent <- GBIF4MIAmaxent %>%
  add_column(Occurrence = "presence")

colnames(GBIF4MIAmaxent) <- c("POINT_X", "POINT_Y", "Occ
urrence")

GBIF4MIAmaxent <- GBIF4MIAmaxent %>%
  dplyr::select(Occurrence, everything())

# Write occurrence data for MaxEnt
write.csv(GBIF4MIAmaxent, "data/GBIF4MIAmaxent.csv", ro
w.names = FALSE)
```

```
# Prepare and fit the MaxEnt model
GBIFPO <- readData(
  occurrence = "data/GBIF4MIAmaxent.csv",
  contEV = "data/ascii_files/",
  maxbkg=20000,
  PA = FALSE
)
```

```
# Plot environmental variables with occurrence points
EV1 <- raster::raster(list.files("data/ascii_files/", fu
ll.names=TRUE)[1])

PO <- read.csv("data/GBIF4MIAmaxent.csv")

plot(EV1, legend=FALSE)
points(PO$POINT_X, PO$POINT_Y, pch = 20, cex = 0.5, col
= "blue")
```

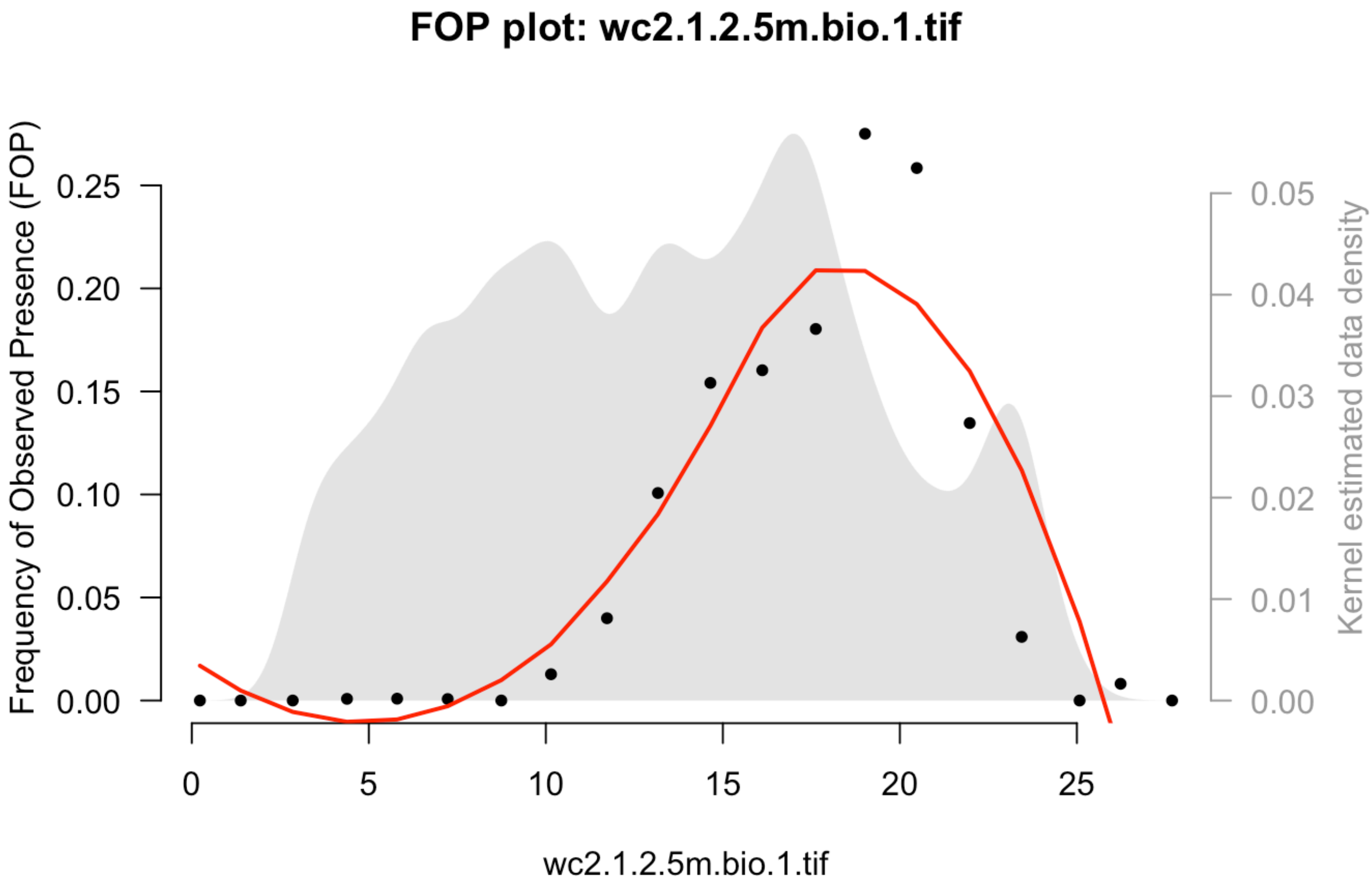


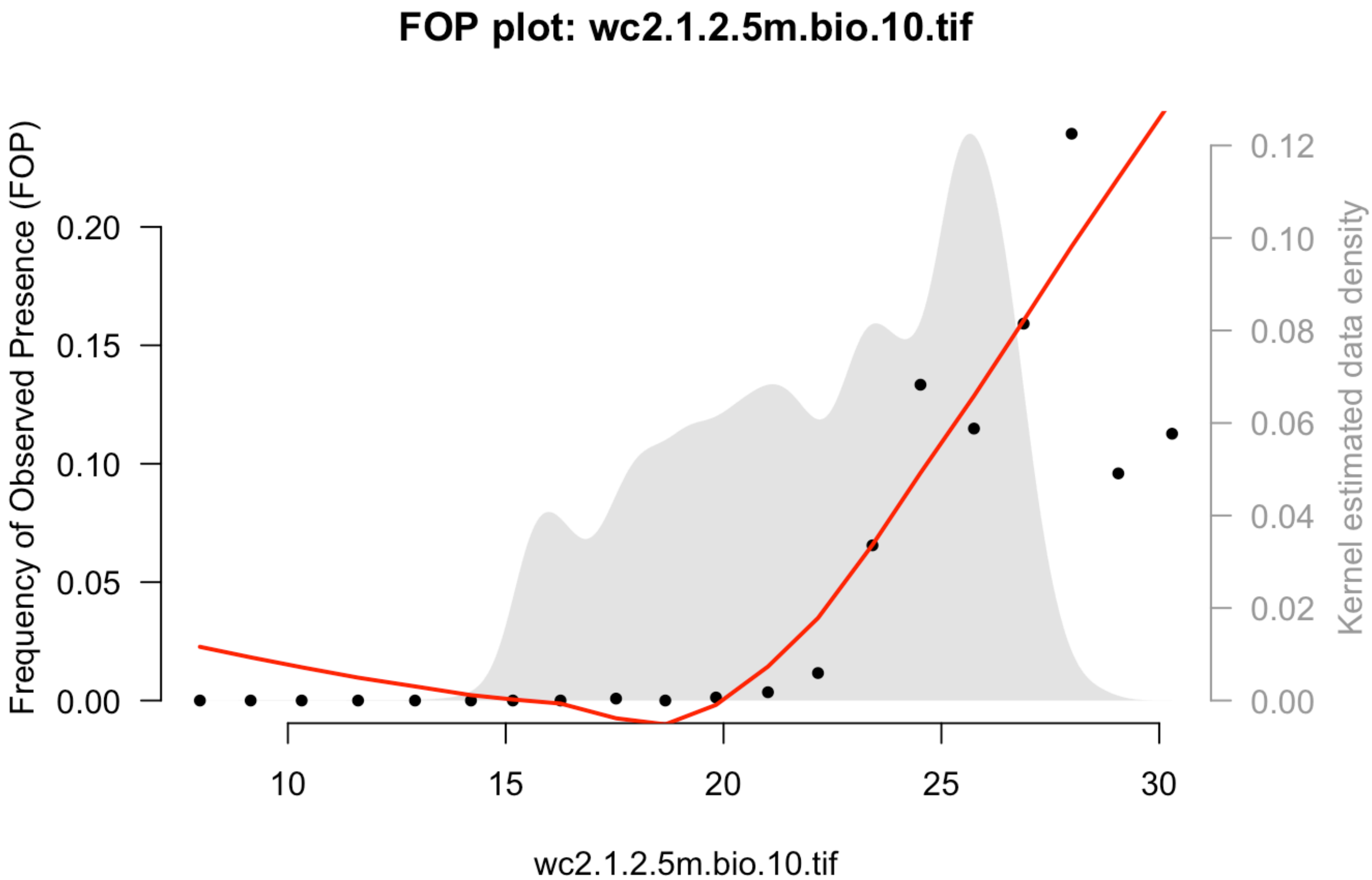

```
str(GBIFPO)
```

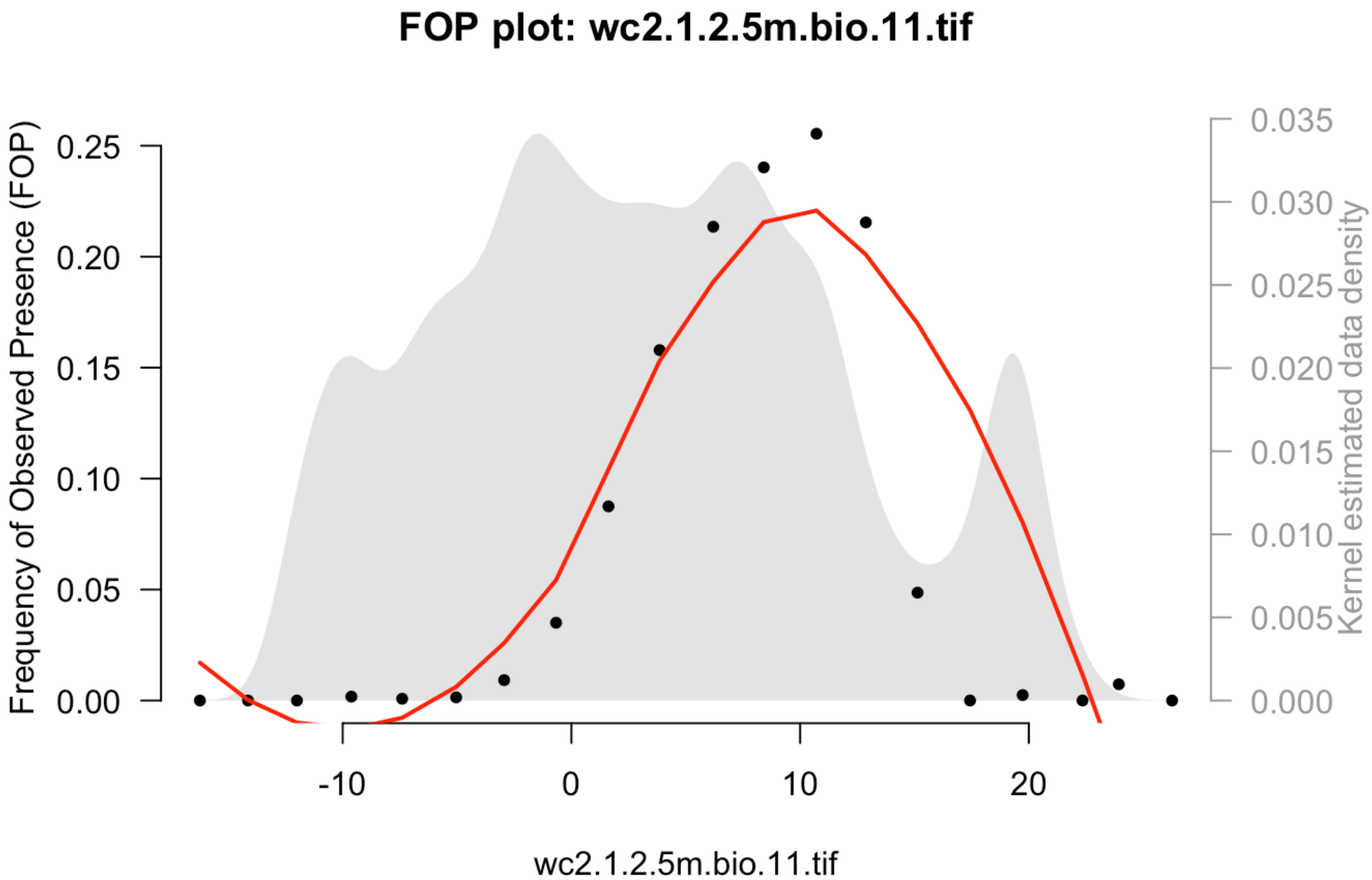
```
## 'data.frame':    21809 obs. of  20 variables:
##  $ RV                 : num  1 1 1 1 1 1 1 1 1 1
...
##  $ wc2.1.2.5m.bio.1.tif : num  15.7 17.8 22.5 20.8 1
5.9 ...
##  $ wc2.1.2.5m.bio.10.tif: num  26.4 26.6 28.9 28.1 2
6.2 ...
##  $ wc2.1.2.5m.bio.11.tif: num  4.21 8.52 15.06 12.81
4.99 ...
##  $ wc2.1.2.5m.bio.12.tif: num  1063 1422 803 1270 118
9 ...
##  $ wc2.1.2.5m.bio.13.tif: num  139 152 145 152 138 11
3 145 163 118 115 ...
##  $ wc2.1.2.5m.bio.14.tif: num  42 85 30 72 64 35 53 8
1 48 47 ...
##  $ wc2.1.2.5m.bio.15.tif: num  31.9 21.3 48.2 24.3 2
0.3 ...
##  $ wc2.1.2.5m.bio.16.tif: num  347 424 321 384 365 27
4 362 421 282 291 ...
##  $ wc2.1.2.5m.bio.17.tif: num  162 263 124 226 255 11
6 192 277 167 152 ...
##  $ wc2.1.2.5m.bio.18.tif: num  248 291 225 371 261 22
6 253 302 184 222 ...
##  $ wc2.1.2.5m.bio.19.tif: num  162 412 134 269 255 11
7 192 394 167 152 ...
##  $ wc2.1.2.5m.bio.2.tif : num  12.75 12.61 9.5 9.61 1
3.44 ...
##  $ wc2.1.2.5m.bio.3.tif : num  34.1 39.7 38.1 36.7 3
6.7 ...
##  $ wc2.1.2.5m.bio.4.tif : num  896 735 566 623 855
...
##  $ wc2.1.2.5m.bio.5.tif : num  33.8 33.1 33.7 33.1 34
...
##  $ wc2.1.2.5m.bio.6.tif : num  -3.58 1.41 8.79 6.88 -
```

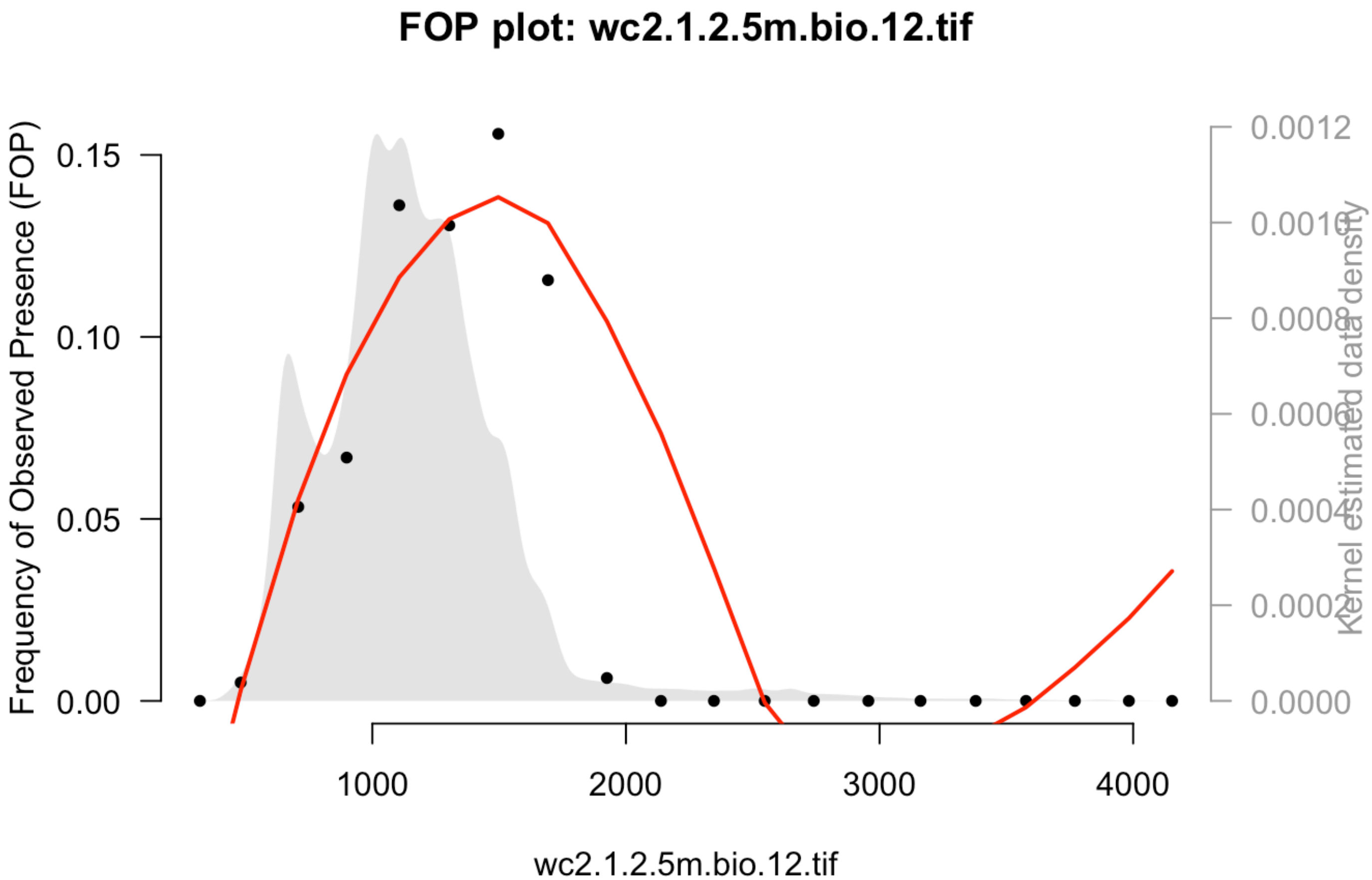
```
2.64 ...
##  $ wc2.1.2.5m.bio.7.tif : num  37.4 31.7 24.9 26.2 3
6.6 ...
##  $ wc2.1.2.5m.bio.8.tif : num  20.1 17.7 27.1 26.8 1
5.6 ...
##  $ wc2.1.2.5m.bio.9.tif : num  4.21 23.16 16.25 17.07
4.99 ...
```

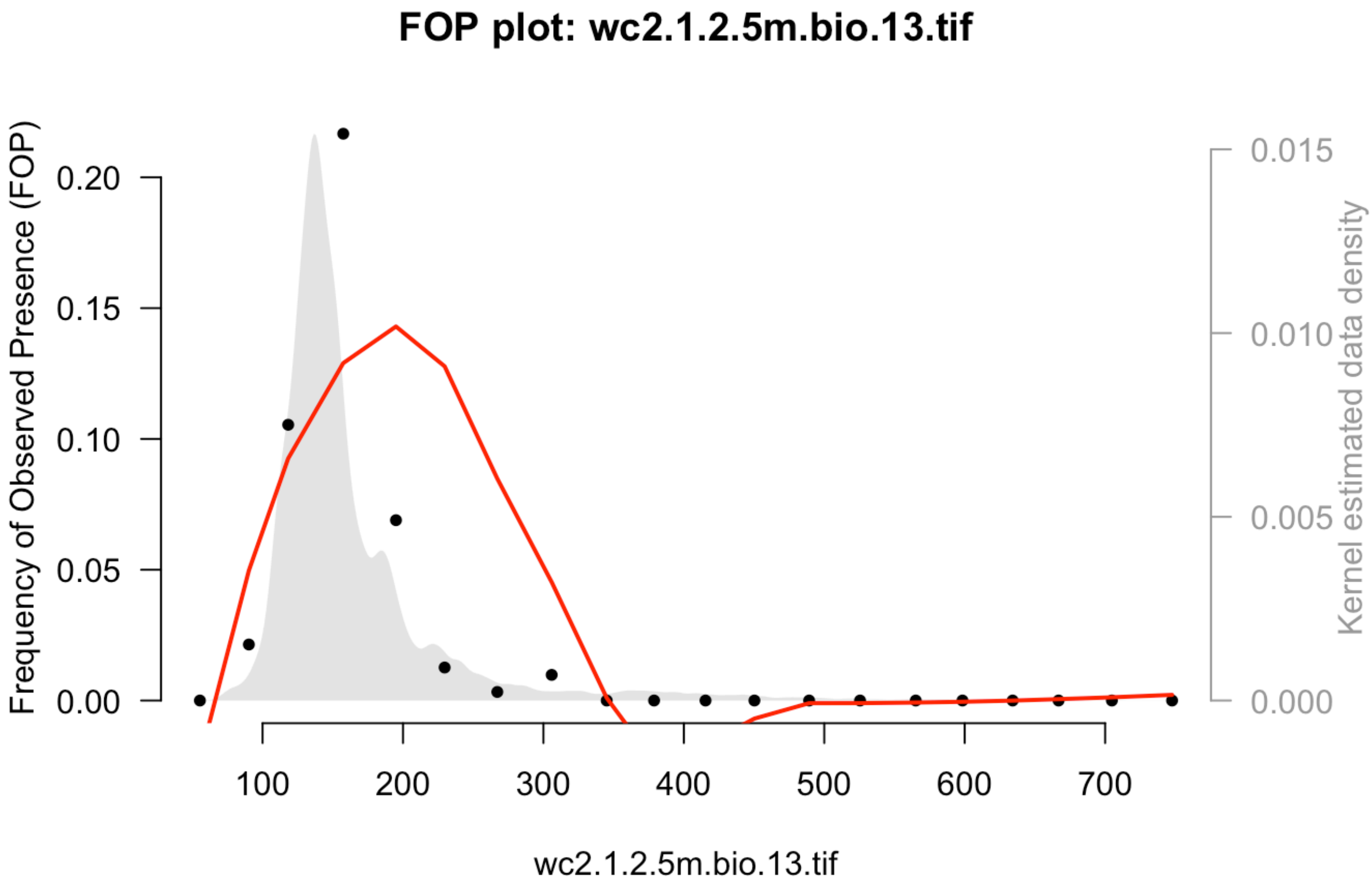
```
for (n in list.files(path = "data/climate/wc2.1_2.5m/",
pattern = ".tif")) {
  plotFOP(GBIFPO, stringr::str_replace_all(n, pattern =
"_", replacement = "."), span = 0.75, intervals = 20)
}
```

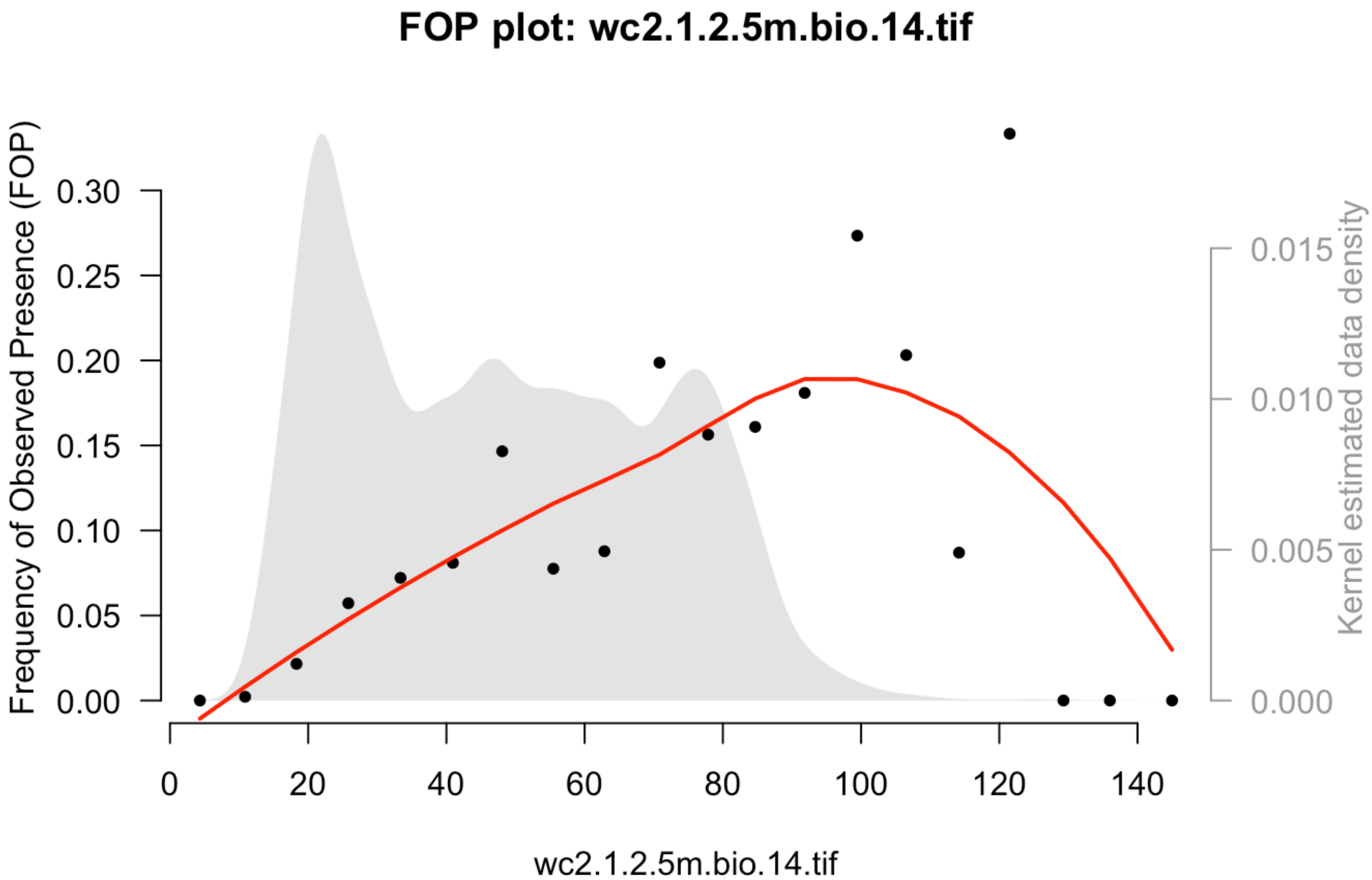

FOP plot: wc2.1.2.5m.bio.1.tif
Frequency of Observed Presence (FOP)
0.25
0.20
0.15
0.10
0.05
0.00
Kernel estimated data density
0.05
0.04
0.03
0.02
0.01
0.00
0
5
10
15
20
25
wc2.1.2.5m.bio.1.tif

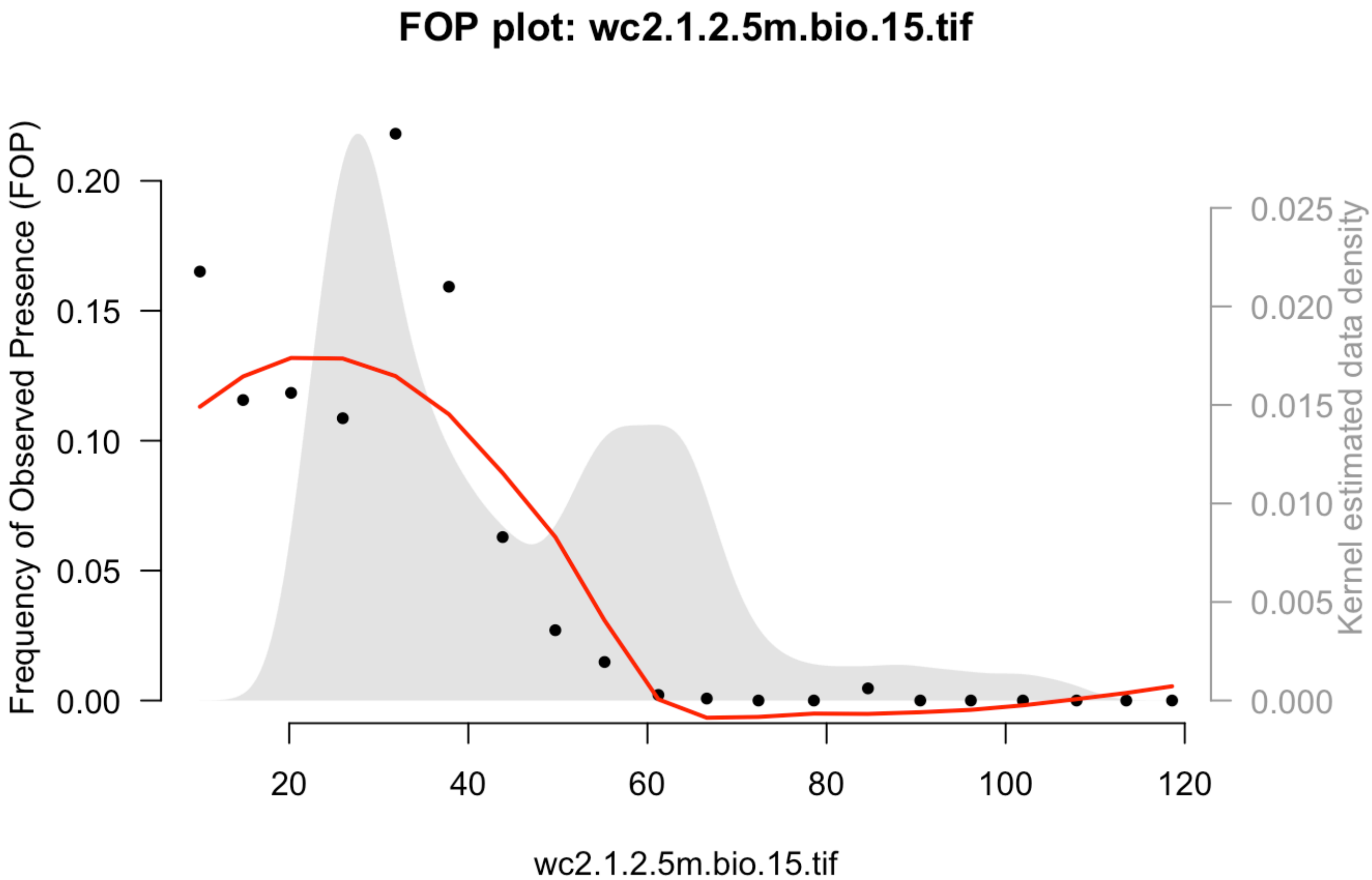

FOP plot: wc2.1.2.5m.bio.10.tif
Frequency of Observed Presence (FOP)
0.00
0.05
0.10
0.15
0.20
Kernel estimated data density
0.00
0.02
0.04
0.06
0.08
0.10
0.12
10
15
20
25
30
wc2.1.2.5m.bio.10.tif

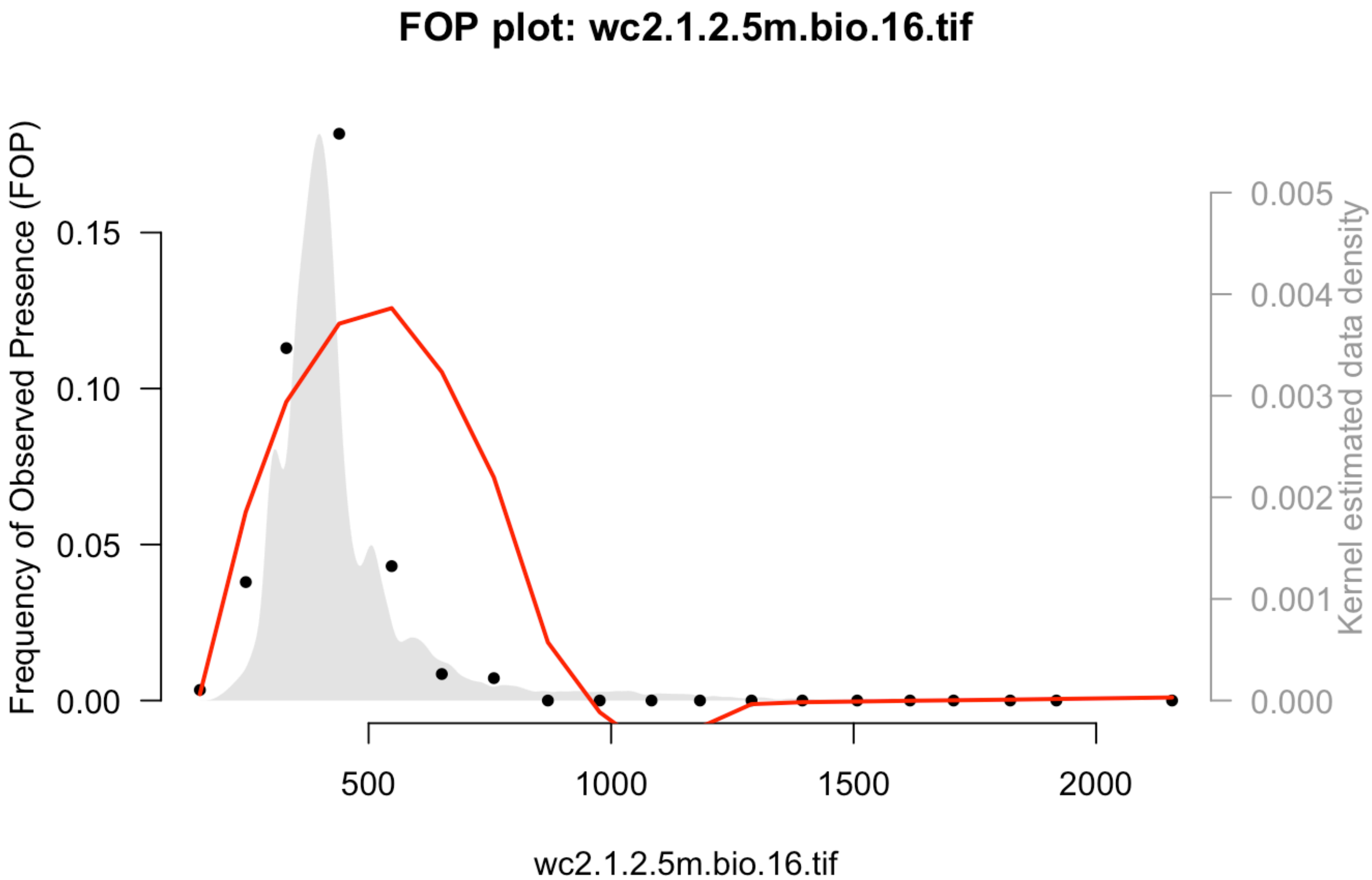

FOP plot: wc2.1.2.5m.bio.11.tif
Frequency of Observed Presence (FOP)
0.25
0.20
0.15
0.10
0.05
0.00
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
Kernel estimated data density
-10
0
10
20
wc2.1.2.5m.bio.11.tif

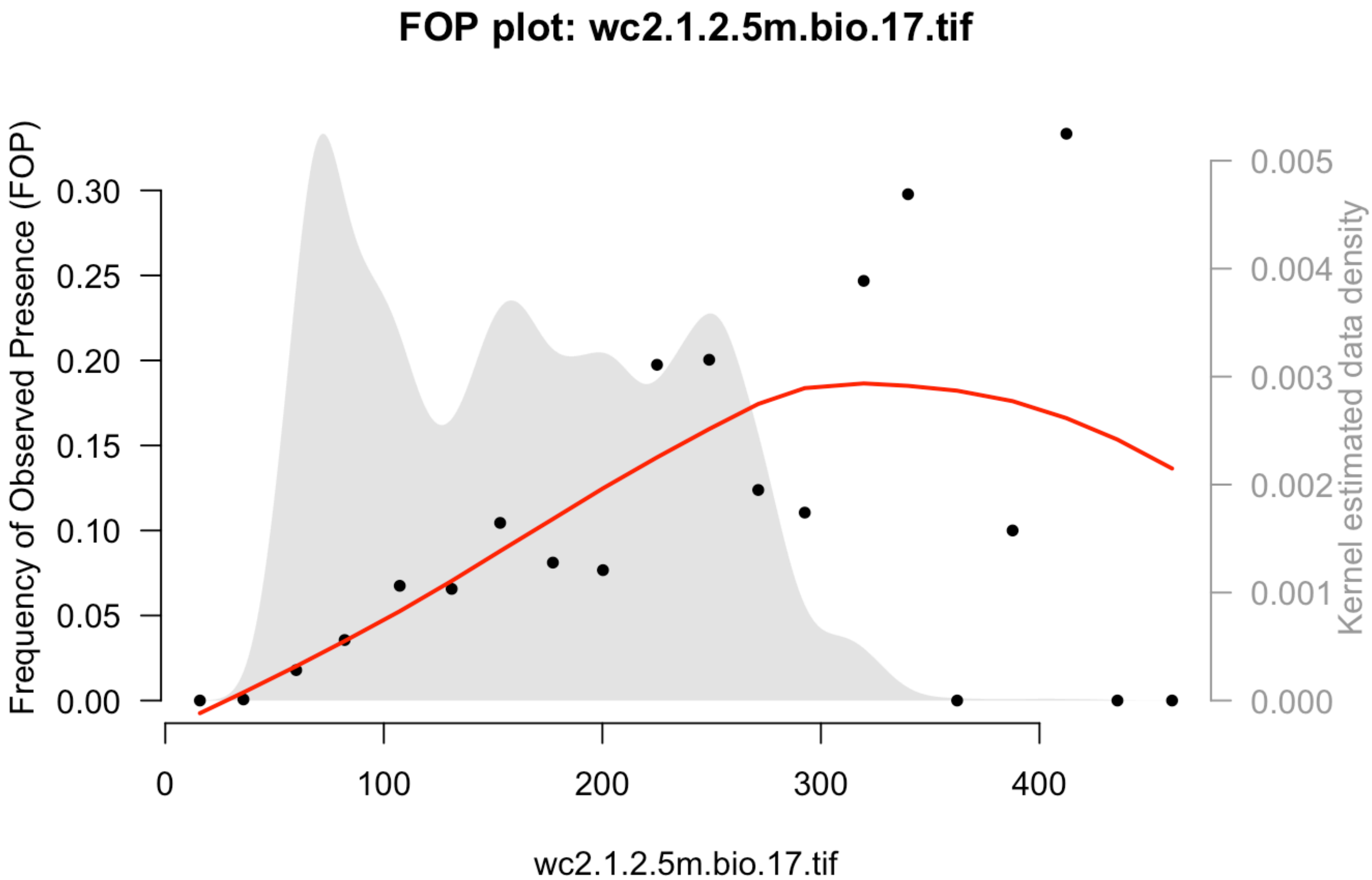

FOP plot: wc2.1.2.5m.bio.12.tif
Frequency of Observed Presence (FOP)
0.15
0.10
0.05
0.00
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002
0.0000
Kernel estimated data density
1000
2000
3000
4000
wc2.1.2.5m.bio.12.tif

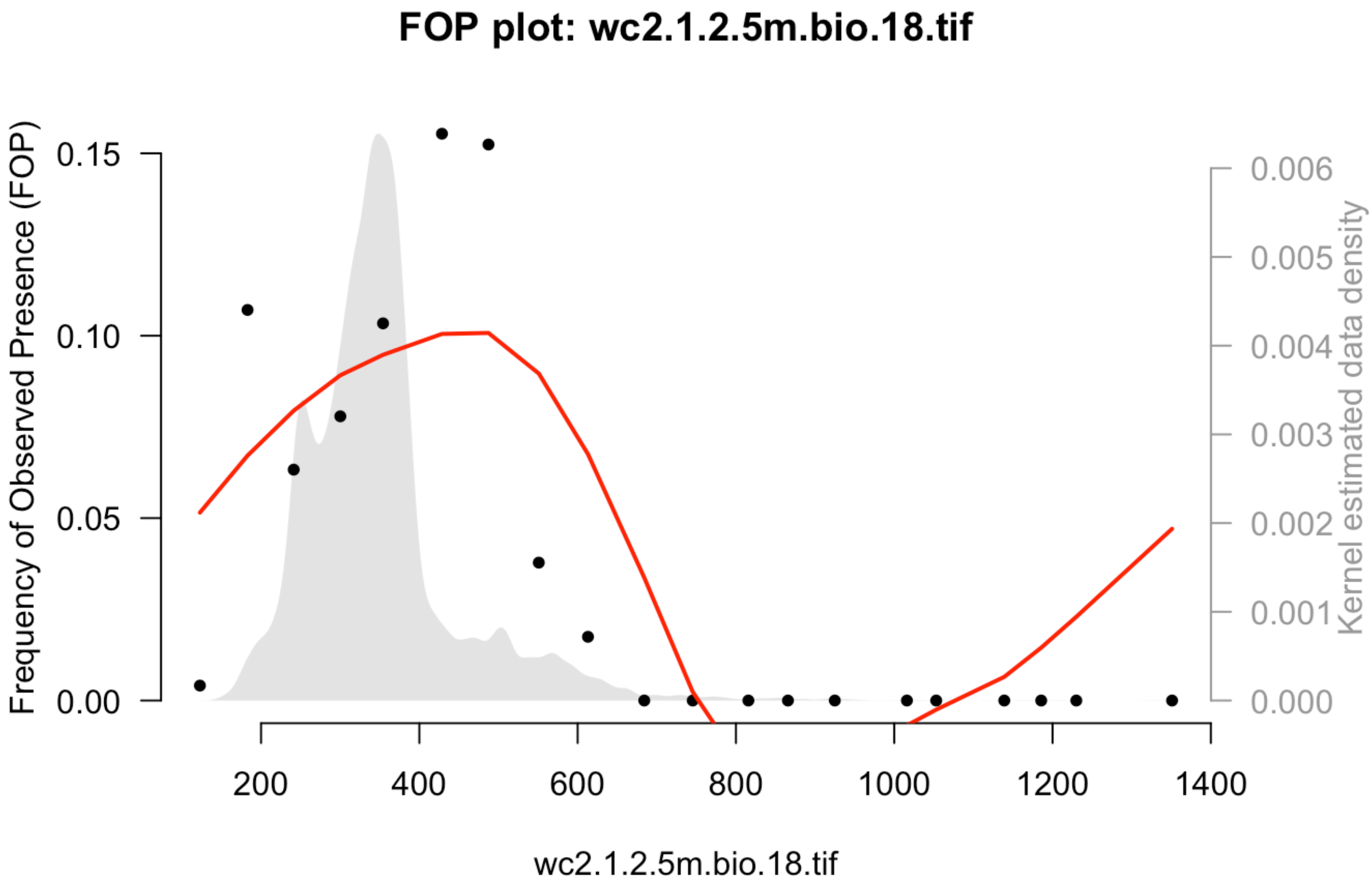

FOP plot: wc2.1.2.5m.bio.13.tif
Frequency of Observed Presence (FOP)
0.20
0.15
0.10
0.05
0.00
Kernel estimated data density
0.015
0.010
0.005
0.000
100
200
300
400
500
600
700
wc2.1.2.5m.bio.13.tif

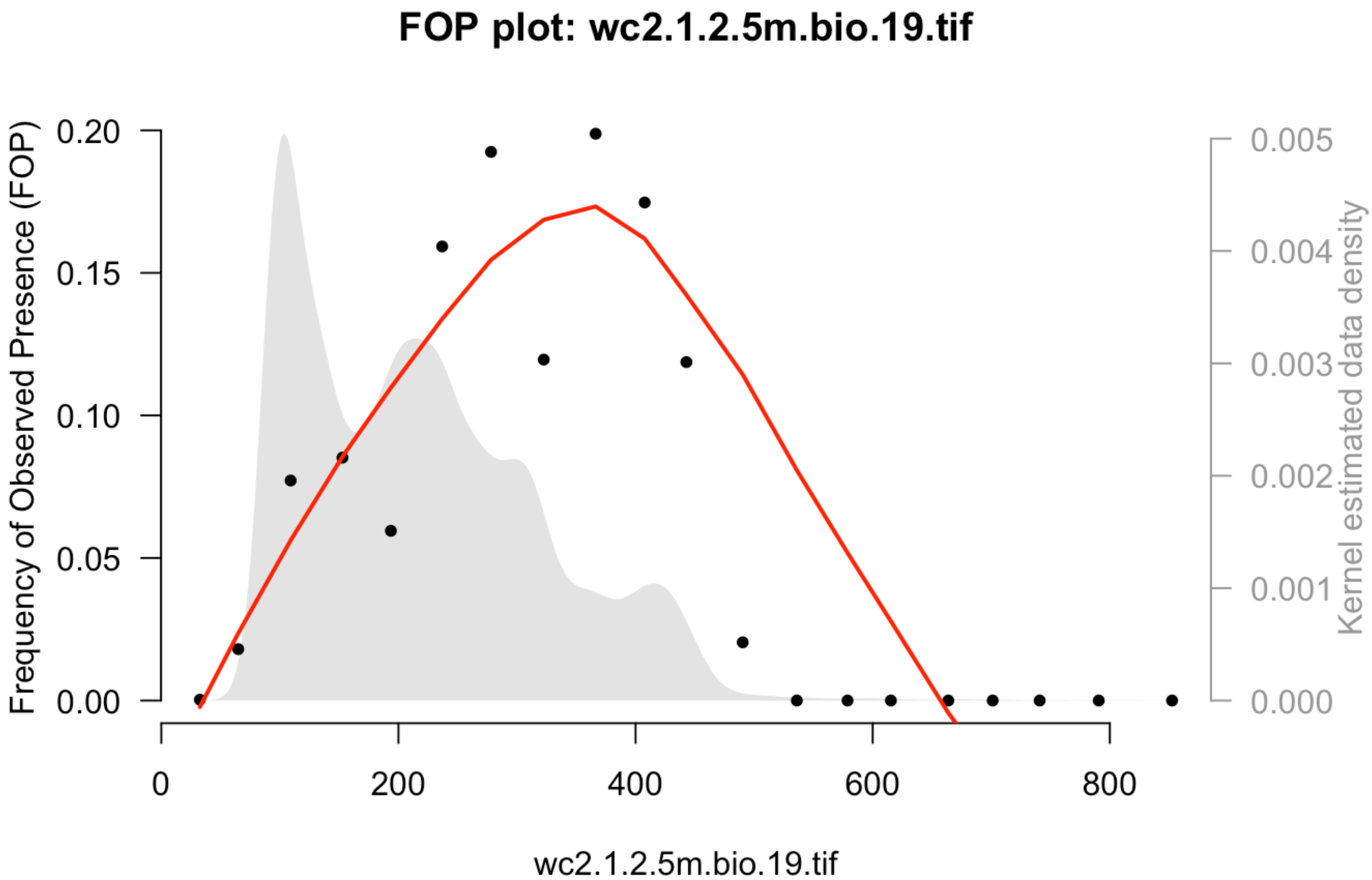

FOP plot: wc2.1.2.5m.bio.14.tif
Frequency of Observed Presence (FOP)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Kernel estimated data density
0.000
0.005
0.010
0.015
0
20
40
60
80
100
120
140
wc2.1.2.5m.bio.14.tif

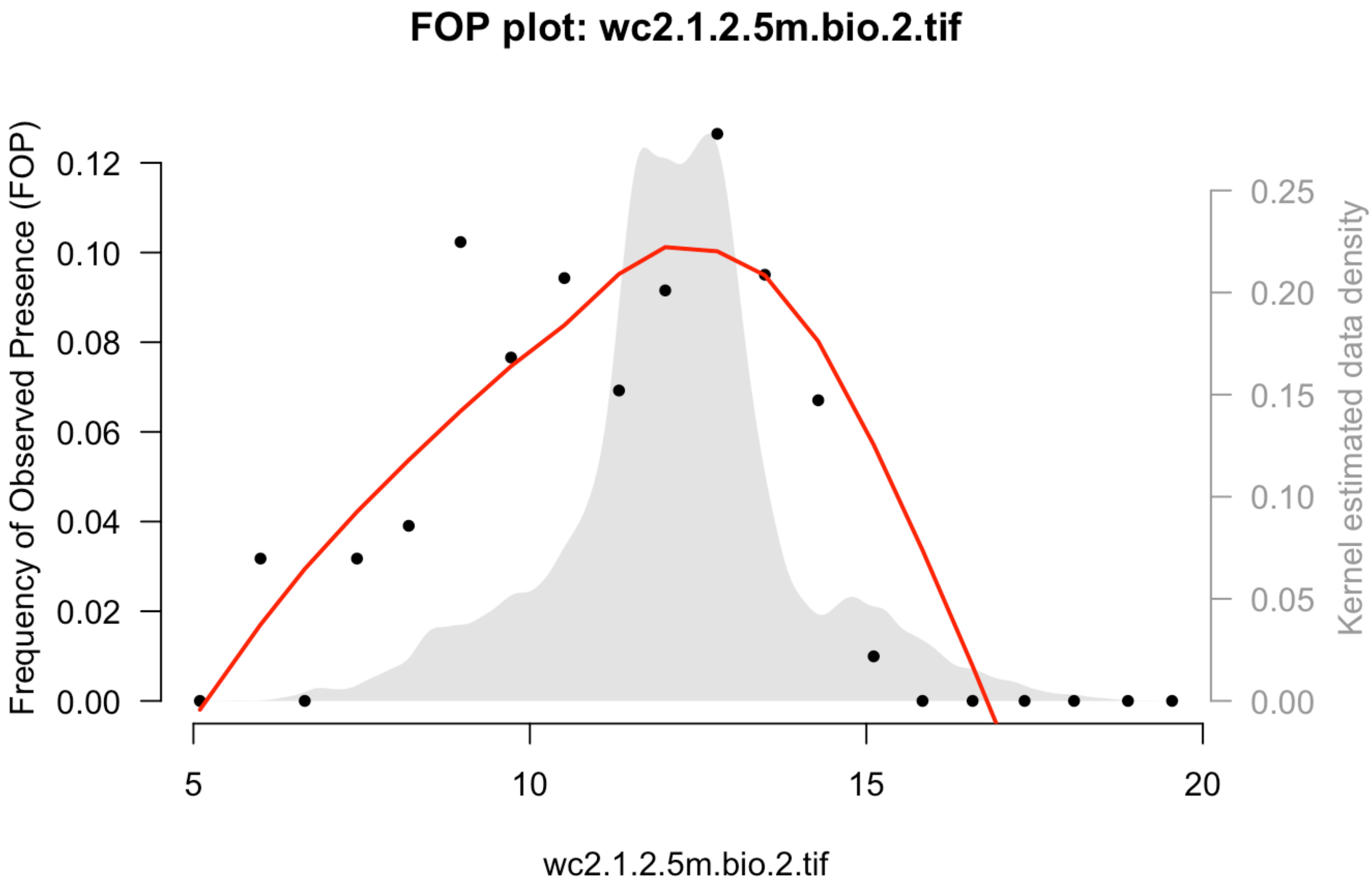

FOP plot: wc2.1.2.5m.bio.15.tif
Frequency of Observed Presence (FOP)
0.20
0.15
0.10
0.05
0.00
Kernel estimated data density
0.025
0.020
0.015
0.010
0.005
0.000
20
40
60
80
100
120
wc2.1.2.5m.bio.15.tif

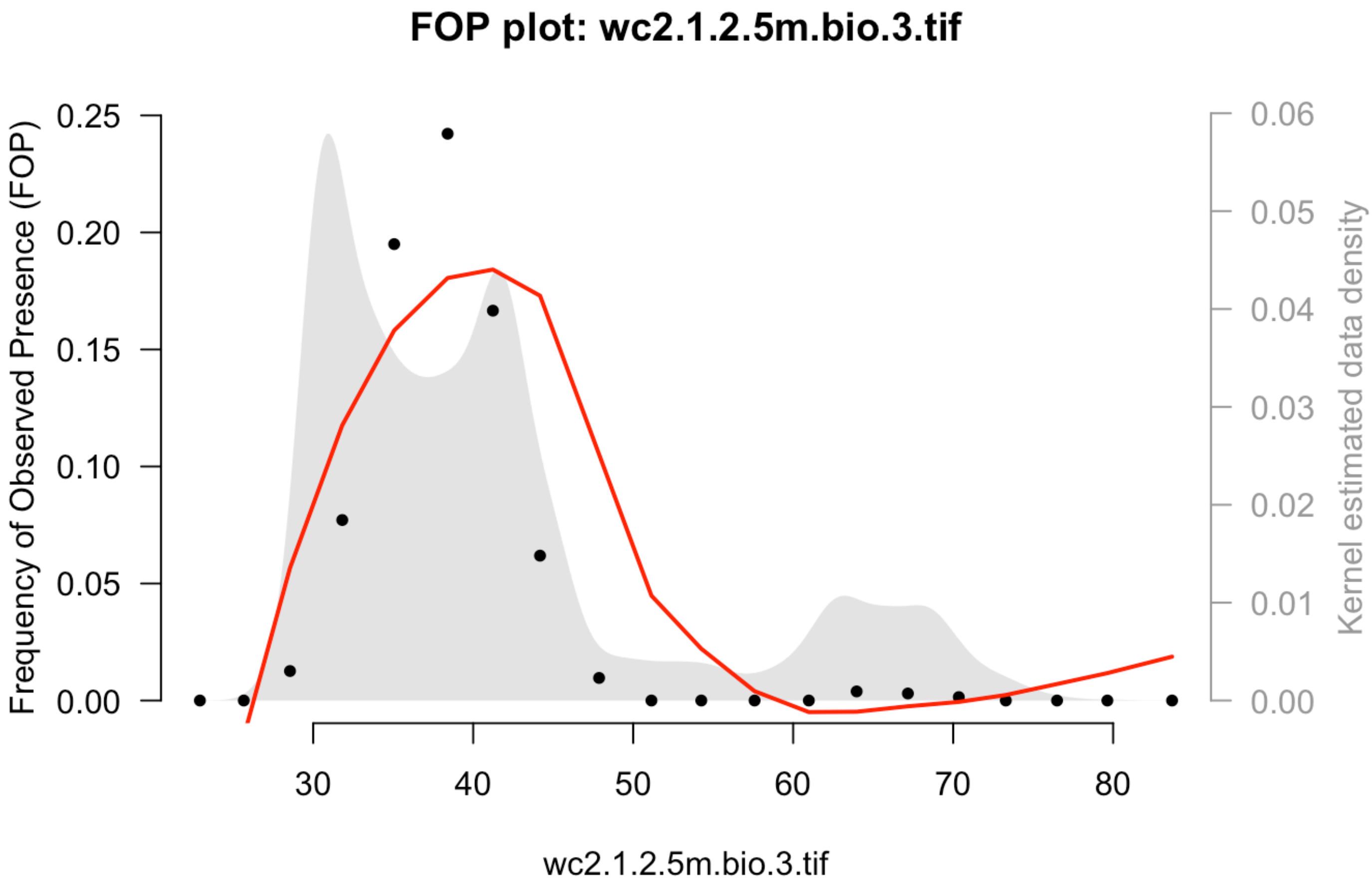

FOP plot: wc2.1.2.5m.bio.16.tif
Frequency of Observed Presence (FOP)
0.15
0.10
0.05
0.00
0.005
0.004
0.003
0.002
0.001
0.000
Kernel estimated data density
500
1000
1500
2000
wc2.1.2.5m.bio.16.tif

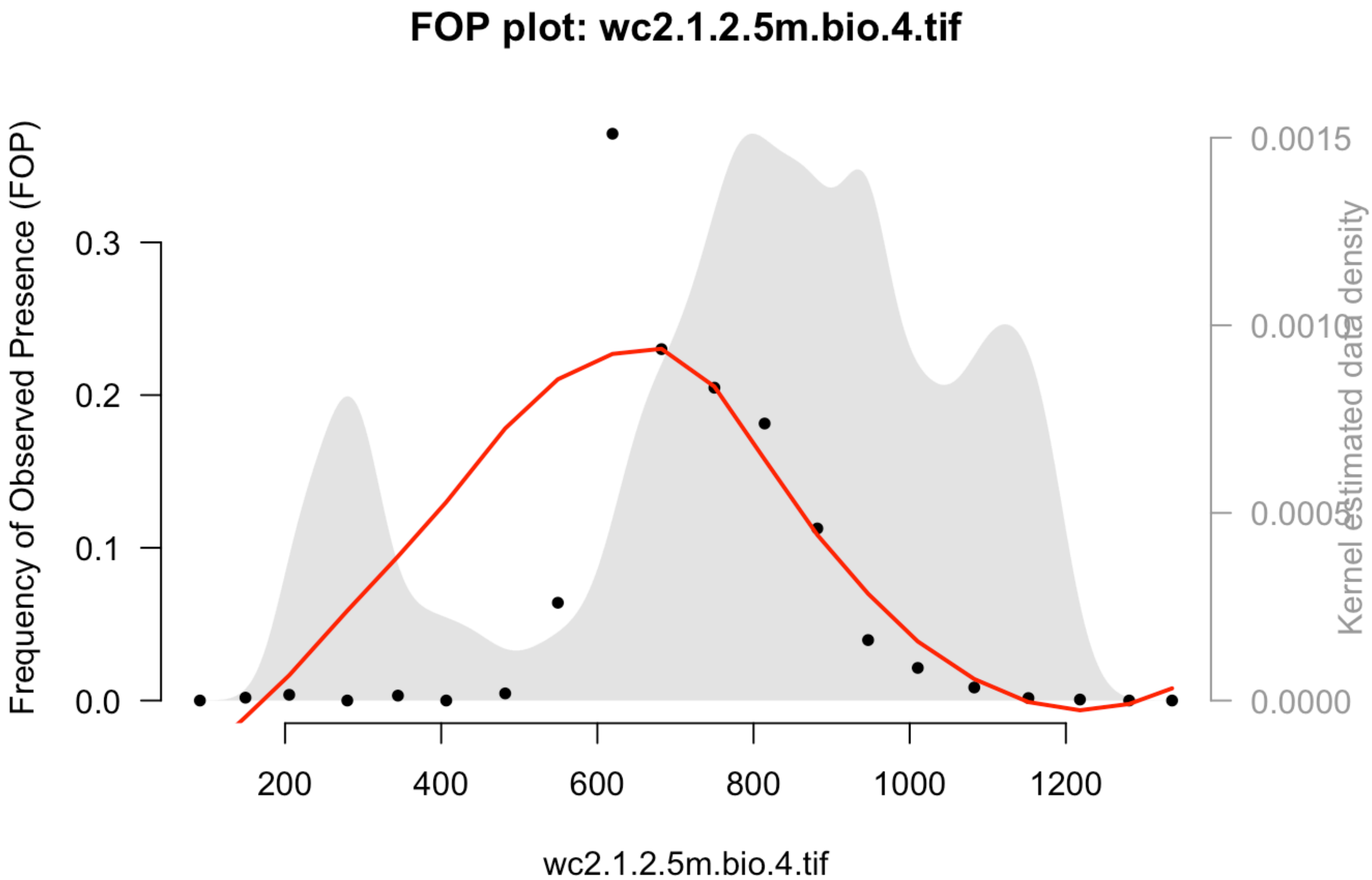

FOP plot: wc2.1.2.5m.bio.17.tif
Frequency of Observed Presence (FOP)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
100
200
300
400
wc2.1.2.5m.bio.17.tif
0.005
0.004
0.003
0.002
0.001
0.000
Kernel estimated data density

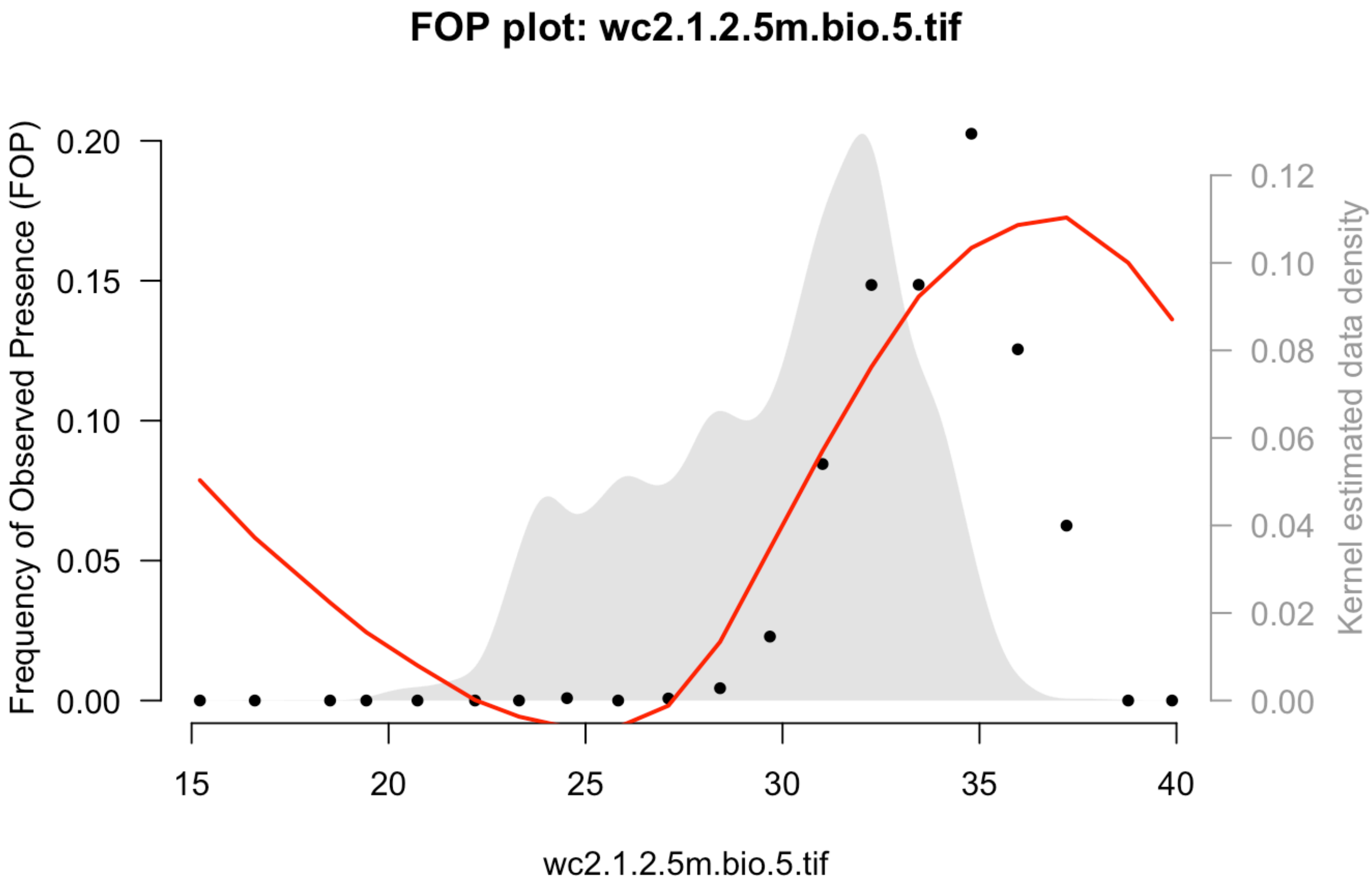

FOP plot: wc2.1.2.5m.bio.18.tif
Frequency of Observed Presence (FOP)
0.15
0.10
0.05
0.00
Kernel estimated data density
0.006
0.005
0.004
0.003
0.002
0.001
0.000
200
400
600
800
1000
1200
1400
wc2.1.2.5m.bio.18.tif

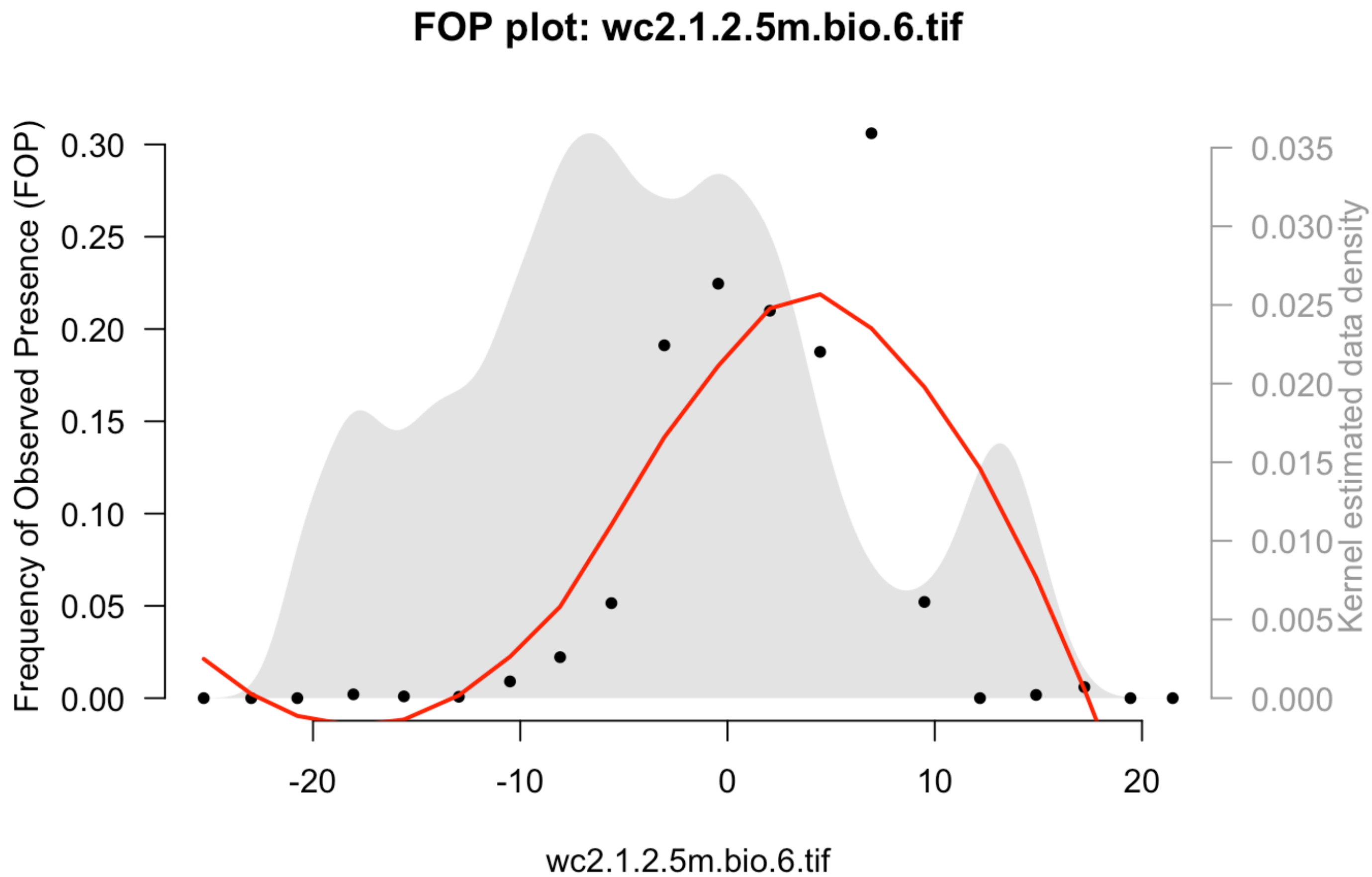

FOP plot: wc2.1.2.5m.bio.19.tif
Frequency of Observed Presence (FOP)
0.20
0.15
0.10
0.05
0.00
Kernel estimated data density
0.005
0.004
0.003
0.002
0.001
0.000
0
200
400
600
800
wc2.1.2.5m.bio.19.tif

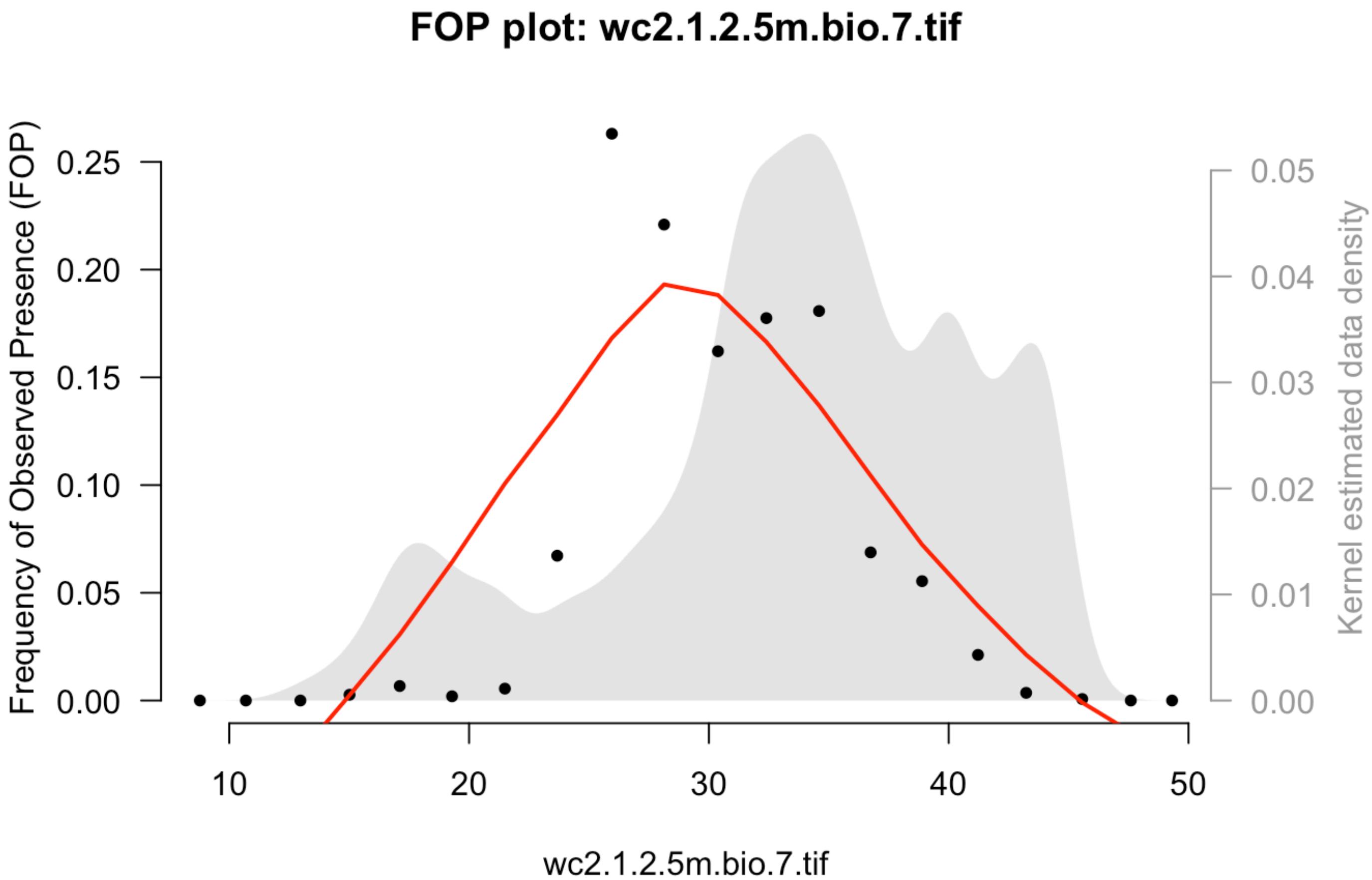

FOP plot: wc2.1.2.5m.bio.2.tif
Frequency of Observed Presence (FOP)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Kernel estimated data density
0.00
0.05
0.10
0.15
0.20
0.25
5
10
15
20
wc2.1.2.5m.bio.2.tif

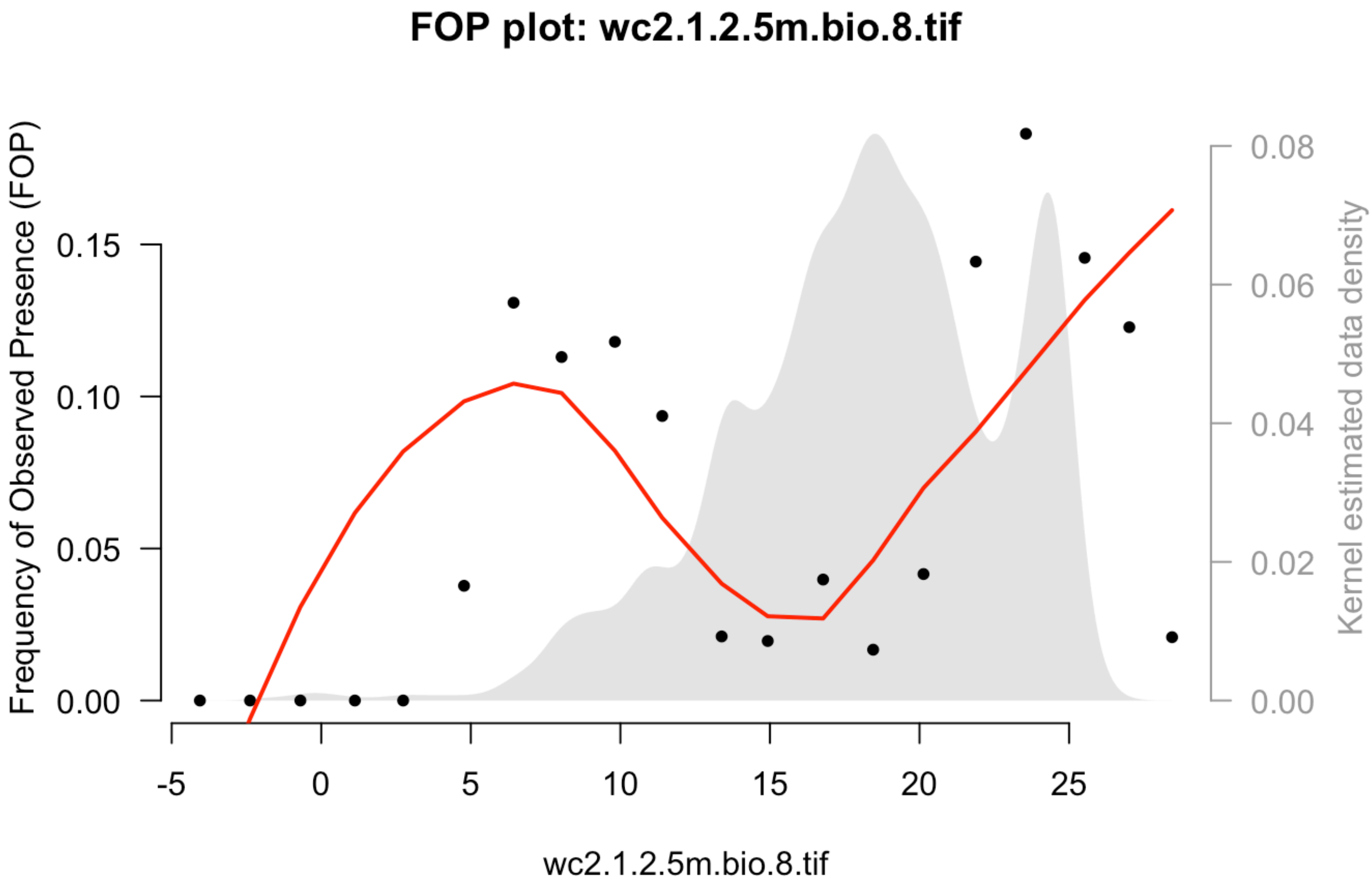

FOP plot: wc2.1.2.5m.bio.3.tif
Frequency of Observed Presence (FOP)
0.25
0.20
0.15
0.10
0.05
0.00
Kernel estimated data density
0.06
0.05
0.04
0.03
0.02
0.01
0.00
30
40
50
60
70
80
wc2.1.2.5m.bio.3.tif

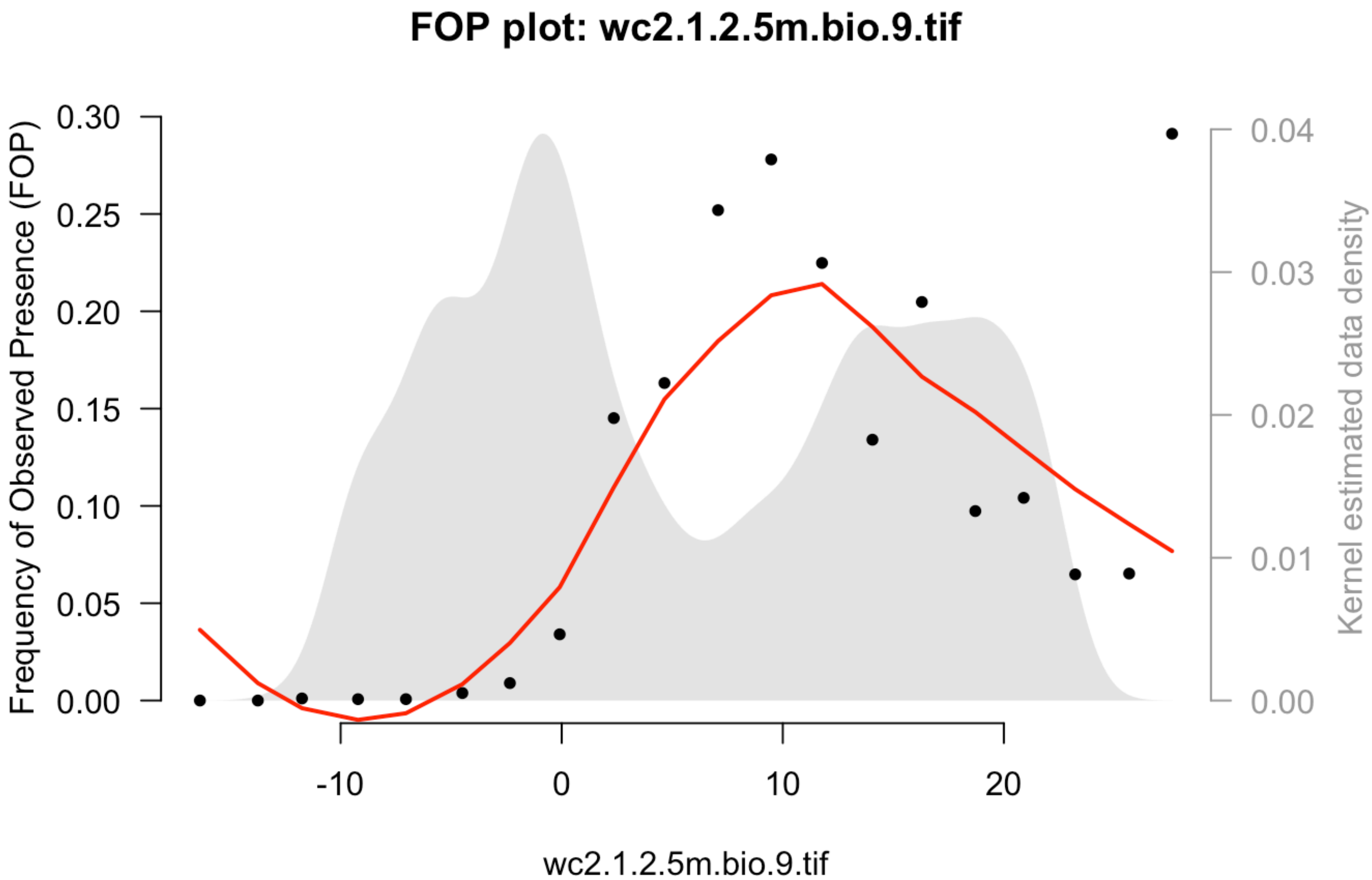

FOP plot: wc2.1.2.5m.bio.4.tif
Frequency of Observed Presence (FOP)
0.0
0.1
0.2
0.3
Kernel estimated data density
0.0000
0.0005
0.0010
0.0015
200
400
600
800
1000
1200
wc2.1.2.5m.bio.4.tif


FOP plot: wc2.1.2.5m.bio.5.tif
Frequency of Observed Presence (FOP)
0.20
0.15
0.10
0.05
0.00
Kernel estimated data density
0.12
0.10
0.08
0.06
0.04
0.02
0.00
15
20
25
30
35
40
wc2.1.2.5m.bio.5.tif


FOP plot: wc2.1.2.5m.bio.6.tif
Frequency of Observed Presence (FOP)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Kernel estimated data density
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
-20
-10
0
10
20
wc2.1.2.5m.bio.6.tif


FOP plot: wc2.1.2.5m.bio.7.tif
Frequency of Observed Presence (FOP)
0.25
0.20
0.15
0.10
0.05
0.00
Kernel estimated data density
0.05
0.04
0.03
0.02
0.01
0.00
10
20
30
40
50
wc2.1.2.5m.bio.7.tif


FOP plot: wc2.1.2.5m.bio.8.tif
Frequency of Observed Presence (FOP)
0.00
0.05
0.10
0.15
Kernel estimated data density
0.00
0.02
0.04
0.06
0.08
-5
0
5
10
15
20
25
wc2.1.2.5m.bio.8.tif



Species Distribution Model - Logistic Regression

```
# Prepare and fit the logistic regression
GBIFDVs.LR <- deriveVars(GBIFPO, algorithm = "LR")
```

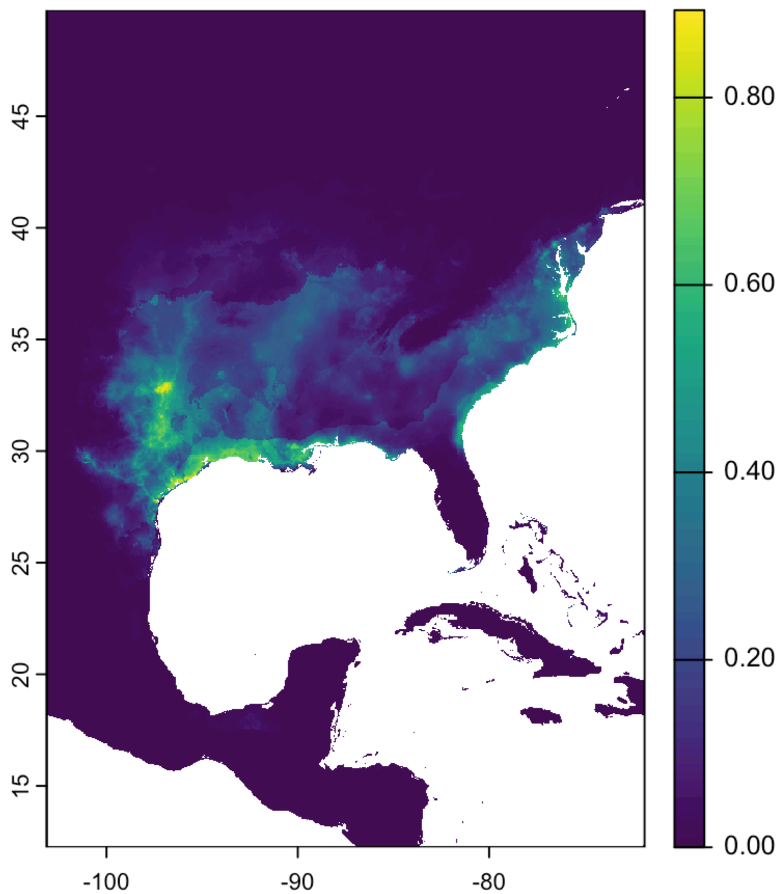
```
GBIFDVselect.LR <- selectDVforEV(GBIFDVs.LR$dvdata, alph
a = 0.001,
                                 algorithm = "LR")
```

```
##   |
|
|   0%  |
|====
|   5%  |
|=======
|  11%  |
|===========
|  16%  |
|===============
|  21%  |
|==================
|  26%  |
|======================
|  32%  |
|==========================
|  37%  |
|=============================
|  42%  |
|=================================
|  47%  |
|=====================================
|  53%  |
|=========================================
|  58%  |
|============================================
|  63%  |
|================================================
|  68%  |
|====================================================
|  74%  |
|=======================================================
|  79%  |
|=======================================================
```

```
====            |  84%  |
|========================================================
========        |  89%  |
|========================================================
===========     |  95%  |
|========================================================
===============| 100%
```

```
GBIFEVselect.LR <- selectEV(GBIFDVselect.LR$dvdata, alph
a = 0.001,
                            algorithm = "LR")
```

```
# Build and project the Logistic Regression model
EVfiles <- c(list.files("data/ascii_files/", pattern =
".asc", full.names=TRUE))
EVstack <- raster::stack(EVfiles)
GBIFPreds.LR <- projectModel(model = GBIFEVselect.LR$sel
ectedmodel,
                               transformations = GBIFDV
s.LR$transformations,
                               data = EVstack)
```



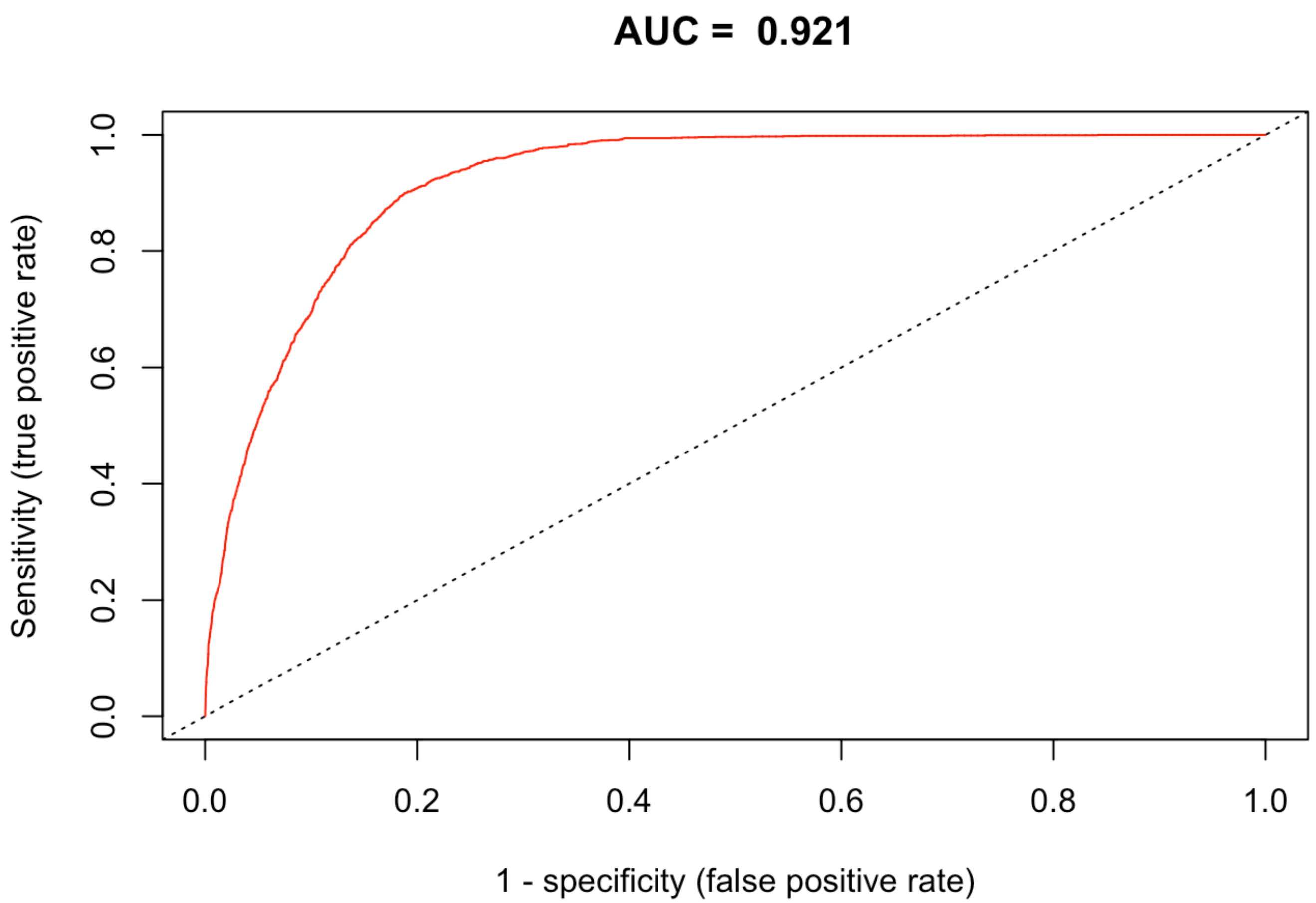

```
# Plot Logistic Regression predictions
pdf("GBIF_MIAMAXENT_Pred_LR.pdf")
plot(GBIFPreds.LR$output)
dev.off()
```

```
## quartz_off_screen
##                  2
```

```
plot(GBIFPreds.LR$output)
```

```
# Test AUC for Logistic Regression model
testAUC(GBIFEVselect.LR$selectedmodel, GBIFDVs.LR$transformations, GBIFPO)
```

```
## Warning: The test data consist of 1/NA only, so NA is
treated as absence.
## Be aware of implications for the interpretation of th
e AUC value.
```

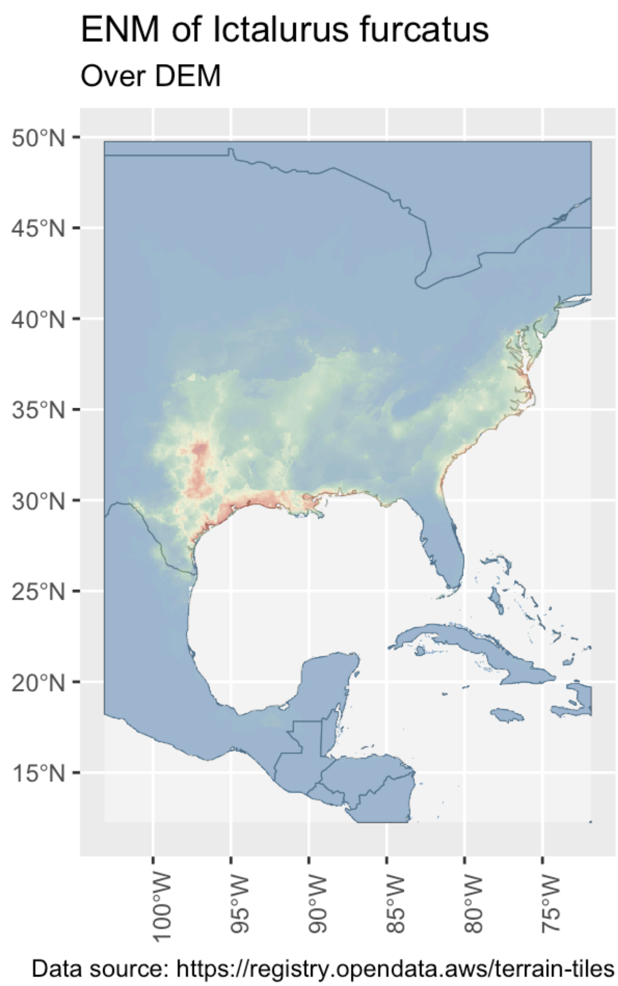



```
## [1] 0.9210379
```

```
sf_use_s2(FALSE)

world <- ne_countries(scale = "medium", returnclass = "s
f")

easternUS <- st_crop(world, c(xmin=-103.125, xmax=-71.87
5, ymin=12.25, ymax=49.75))
```

```
## Warning: attribute variables are assumed to be spatia
lly constant throughout
## all geometries
```

```
ggplot(data=easternUS) +
  geom_sf() +
  geom_spatraster(data = GBIFPreds.LR$output, alpha = 0.
5) +
  scale_fill_whitebox_c(palette = "muted", na.value = "t
ransparent", guide = 'none') +
  labs(title = 'ENM of Ictalurus furcatus',
       subtitle = 'Over DEM',
       caption = 'Data source: https://registry.opendat
a.aws/terrain-tiles') +
  theme(axis.text.x = element_text(angle=90))
```



```
sf_use_s2(TRUE)
```

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- Environmental Data: WorldClim, http://worldclim.org/ (http://worldclim.org/)


```
# Layer citations
print(layer_citations(layercodes))
```

```
## [1] "Domisch S, Amutelli G, Jetz W (2015). \"Near-glo
bal freshwater-specific environmental variables for biod
iversity analyses in 1 km resolution.\"\n_Scientific Dat
a_. doi:10.1038/sdata.2015.73\n<https://doi.org/10.1038/
sdata.2015.73>."
```


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